

Executive summary

Electric car sales continue to break records globally, particularly in China and other emerging economies

Electric car sales exceeded 17 million globally in 2024, reaching a sales share of more than 20%. Just the additional 3.5 million electric cars sold in 2024 compared with the previous year is more than the total number of electric cars sold worldwide in 2020. China maintained its lead, with electric cars accounting for almost half of all car sales in 2024; the over 11 million electric cars sold in China last year were more than global sales just 2 years earlier. As a result of continued strong growth, 1 in 10 cars on Chinese roads is now electric. Europe saw sales stagnate in 2024 as subsidy schemes and other supportive policies waned, but the sales share of electric cars remained around 20% as stronger sales in some countries compensated for lower sales in others. In the United States, electric car sales grew by about 10% year-on-year, reaching more than 1 in 10 cars sold.

Emerging markets in Asia and Latin America are becoming new centres of growth, with electric car sales jumping by over 60% in 2024 to almost 600 000 – about the size of the European market 5 years earlier. In Southeast Asia, electric car sales grew by nearly 50% to represent 9% of all car sales in the region, with notably higher sales shares in Thailand and Viet Nam. In Brazil, the largest car market in Latin America, electric car sales more than doubled to 125 000 in 2024, reaching a sales share of over 6%. Sales in Africa also more than doubled, too, mostly thanks to growing sales in Egypt and Morocco, though electric cars still represent less than 1% of total car sales across the continent. Policy support and relatively affordable electric car imports from China played a central role in increasing sales in some emerging electric vehicle (EV) markets, accounting for 85% of electric car sales in both Brazil and Thailand, for example. Across all emerging economies outside of China, Chinese imports made up 75% of the increase in electric car sales in 2024.

Expected growth in electric car sales is led by China and Europe, alongside a surge in emerging economies

Electric car sales in 2025 are expected to exceed 20 million worldwide to represent more than one-quarter of cars sold worldwide. Sales were up 35% year-on-year in the first three months of 2025, with record first-quarter sales in all major markets. In China, the continuation of incentives for replacing older vehicles and falling electric car prices mean electric cars are projected to reach around

60% of total car sales in the country in 2025. Emissions standards in the European Union and the United Kingdom will require higher shares of zero-emission car sales in 2025. Building on more than 20% year-on-year sales growth observed in the first quarter, these policy pushes are expected to drive up electric car sales in Europe in 2025 to reach a sales share of 25%, despite flexibility given to automakers for meeting the 2025 EU emissions reduction target. While the 2025 outlook for electric car sales in the United States is uncertain based on today's policy direction, sales are currently expected to maintain the 10% growth observed in the first quarter; as consumers take advantage of existing tax credits in view of their potential repeal, electric car sales are projected to reach 11% of total car sales over the full year. In emerging economies other than China, sales are expected to continue growing strongly, increasing by 50% to reach 1 million in 2025.

Despite uncertainties in the outlook, the share of electric cars in overall car sales is set to exceed 40% in 2030 under today's policy settings. China is poised to continue leading in electric car sales to 2030, achieving a sales share of around 80% on the back of significant market momentum and competitively-priced EVs. In Europe, carbon dioxide (CO₂) targets support the achievement of a sales share of close to 60% by 2030, slightly below last year's projection. The sales share in the United States grows much more modestly than in our *Outlook* last year, reaching around 20% by 2030 based on today's policy direction – less than half the share projected for 2030 last year. In Southeast Asia, meanwhile, electric car sales are boosted by strong policy support and available domestic manufacturing capacity: by 2030, one in four cars sold in the region is poised to be electric. Two/three-wheelers – an important mode of road transport in the region – electrify faster: by 2030, almost 1 in 3 such vehicles sold are electric. Across all vehicle modes, the deployment of EVs replaces the use of more than 5 million barrels of oil per day globally in 2030, an important energy security consideration. Half of these savings are the result of EV adoption in China.

Uncertainty about the evolution of trade and industrial policy, downside risks to the economic outlook, and lower oil prices could affect EV uptake – but also car markets overall. Higher tariffs might increase the price of cars, including electric cars, and their components; lower GDP growth could dampen car sales; and lower oil prices affect the fuel cost savings from the use of electric cars. The way these factors will play out in practice is uncertain, but on aggregate they look to pose risks for overall car sales volumes more than for the share of EVs. In China, continued political support and competitive EV prices suggest that EV sales can withstand such headwinds. In Europe, where price differentials with conventional cars are larger, the combination of longer-term policy ambition and examples of policy responses from the pandemic suggest that sustaining EV sales is possible. Low oil prices can reduce fuel savings offered by battery electric cars, but even at global benchmark oil prices of USD 40 per barrel, there would be

significant fuel cost savings in all major EV markets when charging at home. In China, public fast charging costs around twice as much as home charging, but it would still offer savings for EV owners compared with driving a conventional car.

Global trade of electric cars is growing as manufacturers eye new markets

China continues to be the world's EV manufacturing hub and is responsible for more than 70% of global production. Car manufacturers headquartered in China predominantly cater to the domestic market, accounting for around 80% of domestic sales in 2024 and almost all of the 25% growth in global EV production. In the European Union, production stalled at 2.4 million electric cars in 2024. North America saw contrasting trends: US production declined whereas Mexico's output doubled, supported by comparatively low manufacturing costs. Roughly 70% of Mexico's output was from US-headquartered manufacturers. Production also increased by 15% in Asia Pacific countries other than China, reaching about 1 million electric cars, mostly from incumbent carmakers from Japan and Korea.

Global trade of electric cars increased 20% in 2024; imports now represent almost one-fifth of global electric car sales. At 40% (nearly 1.25 million electric cars), China accounted for the largest share of global exports in 2024. The European Union also remained a net exporter of electric cars: exports reached over 800 000, mostly destined for other European countries (such as the United Kingdom) and North America. EU imports were below 700 000, of which 60% came from China. The United States remained a net importer of electric cars; imports increased by nearly 40% in 2024, while exports fell by nearly 15%.

Chinese EV export markets are diversifying as Chinese automakers make headway in Brazil, Mexico and Southeast Asia. Several potential markets for Chinese EV exports have recently adopted or are considering the use of tariffs. This has prompted Chinese manufacturers to either frontload their exports before such tariffs come into force (as in Brazil) or to seek new markets for their available output. Today, overseas manufacturing capacity from Chinese manufacturers supplies about 5% of electric car sales in emerging markets and is set to grow further.

Competition and declining battery prices are improving affordability, though progress is uneven

As a global average, the price of battery electric cars fell in 2024, but the purchase price gap with conventional cars persisted in many markets. The average battery electric car price in Germany, for example, remained 20% higher than that of its conventional counterpart. In the United States, battery electric cars remained 30% more expensive, dampening future sales growth expectations. In

contrast, two-thirds of all electric cars sold in China in 2024 were priced lower than their conventional equivalents, without considering purchase incentives for EVs. This helped boost sales even as government incentives decreased.

Chinese electric car models are typically cheaper than the average EV in emerging markets, bolstering the competitive position of the Chinese industry. In Thailand, the average price of a battery electric car has now reached parity with an average conventional car, and the Chinese electric cars available are, on average, even cheaper. In Brazil, the price gap between battery electric cars and conventional cars shrank to 25% in 2024 from over 100% in 2023 as Chinese electric car imports grew to 85% of the country's EV sales. Similarly, the average price premium of battery electric cars in Mexico fell to 50% in 2024 from more than 100% in 2023 as Chinese imports reached two-thirds of sales.

Low critical mineral prices and increasing competition between battery manufacturers drove down battery pack prices in all markets in 2024, albeit with significant variations. In China, prices fell by about 30%, compared with 10-15% in Europe and the United States. The faster pace of cost reductions in China – enabled by strong competition, increasing manufacturing efficiency and supply chain integration, and access to a skilled workforce – is increasing the competitive advantage of Chinese battery manufacturers.

Improving charging coverage, capacity and integration is key to growing adoption of electric vehicles

Public charging stations have doubled in the past 2 years to keep up with growing EV sales. China and the European Union have maintained a steady pace of charger deployment compared with the number of EVs on the road. However, in both the United States and United Kingdom, which have higher rates of access to home chargers than China, public charger build-out has not kept pace with EV deployment, and the number of electric light-duty vehicles per public charging point increased in 2024.

Charging capacity is an important measure of the adequacy of public charging networks. The number of ultra-fast chargers, with power ratings of 150 kilowatts (kW) and above, grew by about 50% in 2024 and now account for nearly 10% of all public fast chargers. Public slow chargers in urban areas are a solution for EV owners without access to home charging, but fast chargers along highways help enable long-distance trips. In Europe, over three-quarters of all highways have a fast-charging station at least every 50 kilometres, compared with less than half of US highways. Globally, public charging capacity for light-duty EVs would need to grow by almost ninefold to 2030 to support EV sales implied by stated policies. Even so, EVs are set to account for only 2.5% of total global electricity demand in 2030.

Government efforts to promote interoperability, standardisation, smart charging and vehicle-to-grid integration can ease the transition to EVs for drivers and grid operators. Technologies such as smart chargers and vehicle-to-grid-capable EV models are becoming more available, but new market structures and legal frameworks will be needed for vehicles to maximise the potential benefits for the grid. China and the United Kingdom stand out for policy implementation and demonstration projects in this area. Battery technology innovations in recent years are also enabling safe high-power charging that could be as quick as refuelling a conventional car, but fully realising these advances hinges on deploying adequate infrastructure.

The value proposition for electric trucks is improving, even for long-haul operations

Electric truck sales grew by nearly 80% globally in 2024 to reach close to 2% of total truck sales. Spurred by a new scrappage scheme for conventional trucks, China's electric truck sales doubled to 75 000 vehicles, representing over 80% of global sales in 2024. In Europe and the United States, electric truck sales in 2024 remained similar to 2023 levels. The number of battery electric truck models available has grown from less than 70 in 2020 to more than 400, increasing the number of applications that can be met by electric trucks.

The total cost of ownership of a battery electric heavy-duty truck is already lower than for a diesel equivalent in China in certain cases. The purchase price of battery electric heavy-duty trucks remains two to three times higher than equivalent diesel trucks in major markets, which can discourage fleet operators, who often operate under tight margins. However, greater efficiency and lower energy costs – even including the cost of a high-power charger – make battery electric trucks more attractive the more they are used. By 2030, battery electric trucks in Europe and the United States are expected to reach parity on the total cost of ownership with diesel trucks for long-haul operations, as they already have in China, and are set to remain more cost-effective than hydrogen fuel cell trucks. However, the specific application and use profile of trucks are key factors for determining which powertrain technology is most suitable, meaning case-by-case analysis may be needed to evaluate the costs of various alternatives.

Mandated rest periods for truck drivers can play an important role in reducing the opportunity cost of charging en route. For long-haul trucking, the wait time for recharging an electric truck can pose difficulties for logistics operators. However, the 45-minute rest period mandated in the European Union can accommodate the addition of around 150 km of driving range for a heavy-duty truck using a 350 kW charger, and up to about 400 km if a megawatt charger is used. While the United States and China also have mandated rest periods, the design of the EU policy is currently most conducive to supporting EV adoption.

1. Trends in electric car markets

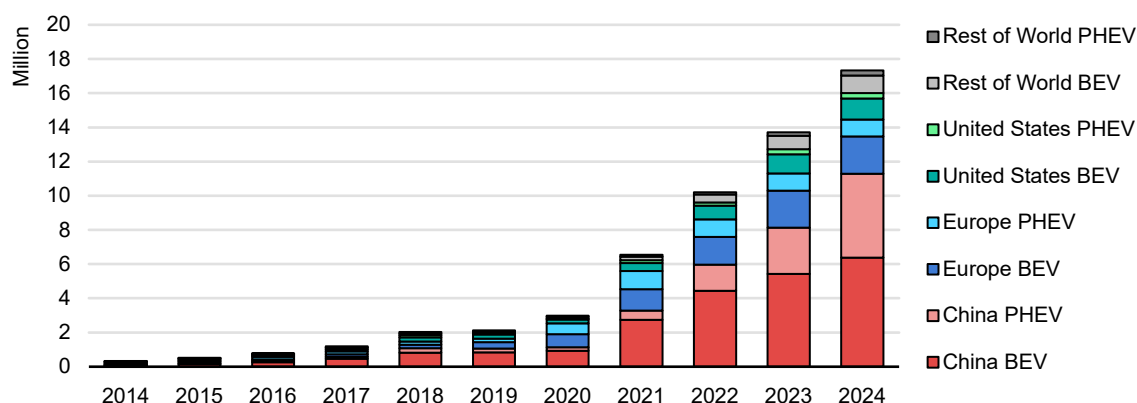
Electric car sales

Global electric car sales exceeded 17 million in 2024

More than 20% of new cars sold worldwide were electric

Electric car sales topped 17 million worldwide in 2024, rising by more than 25%.¹ Just the additional 3.5 million cars sold in 2024 compared to 2023 outnumber total electric car sales in the whole of 2020. China maintained its lead among major markets, with electric car sales exceeding 11 million – more than were sold worldwide just 2 years earlier. Global sales were slightly tempered by stagnating growth in Europe, as subsidies were phased out or reduced in several major markets, and as the EU CO₂ targets for cars remained the same between 2023 and 2024. Electric car sales continued to increase in the United States although growth was about one-quarter that of the previous year. Significantly, outside of these three major markets, there was a record increase in sales of nearly 40% to reach 1.3 million, closing in on the United States' sales of 1.6 million electric cars.

Global electric car sales, 2014-2024



IEA. CC BY 4.0.

Notes: BEV = battery electric vehicle; PHEV = plug-in hybrid vehicle. Includes new passenger cars only.

Sources: IEA analysis based on country submissions and data from the European Automobile Manufacturers Association (ACEA), European Alternative Fuels Observatory (EAFO), EV Volumes and Marklines.

¹ In this report “sales” represents an estimate of the number of new vehicles hitting the roads. Where possible, data on new vehicle registrations is used. In some cases, only data on retail sales (such as sales from a dealership) are available. New car sales or registrations exclude used cars. Unless otherwise specified, the term electric vehicle is used to refer to both battery electric and plug-in hybrid electric vehicles but does not include fuel cell electric vehicles. For a brief description of the trends related to fuel cell electric vehicles, see the box at the end of [Section 4](#).

The rapid growth in electric car sales over the past 5 years has had a significant impact on the global car fleet: At the end of 2024, the electric car fleet had reached almost 58 million, about 4% of the total passenger car fleet and more than triple the total electric car fleet in 2021. Notably, the global stock of [electric cars displaced](#) over 1 million barrels per day of oil consumption in 2024. Of course, the stock of electric cars is not spread evenly across the world – in China, for example, around one in ten cars on the road is now electric, whereas in Europe the ratio is closer to one in twenty.

Almost half of China's car sales were electric in 2024, representing almost two-thirds of electric cars sold globally

Electric car sales in China increased by almost 40% year-on-year in 2024, further driving up China's share of global electric car sales. In 2021, China accounted for half of global electric car sales; this share grew to almost two-thirds in 2024. On a monthly basis, sales of electric cars have overtaken conventional car sales in the country since July 2024, bringing the share of electric car sales close to 50% for the full year. In China, 2024 marks the fourth consecutive year in which the electric car sales share grew by approximately 10 percentage points year-on-year.

The growth in China reflects in no small part the [growing price competitiveness](#) of battery electric cars with conventional cars in the country. In addition, China's electric car market benefitted from the introduction of a [trade-in scheme](#) in April 2024. The scheme, which is part of a wider economic stimulus package, applies to the purchase of conventional and electric cars alike, but with different levels of financial support. Following a [retroactive increase](#), the scheme offers CNY 20 000 (Yuan renminbi) (USD 2 750) for consumers that replace an older vehicle (conventional or electric) with a new electric car, and CNY 15 000 (USD 2 050) for replacement with a new conventional vehicle. In 2024, about [6.6 million](#) consumers applied for the incentive, 60% of whom bought an electric car. As such, more than one-third of the over 11 million new electric car sales in the country benefitted from this incentive.

In recent years, sales of plug-in hybrid electric cars have been growing faster than sales of battery electric cars in China. The share of plug-in hybrid electric vehicle (PHEV) sales, excluding extended-range EVs (EREVs)², in China's total electric car sales has risen from about 15% in 2020 to nearly 30% in 2024. Meanwhile, the share of EREVs has more than quadrupled since 2020, surpassing 10% in

² If not otherwise specified, PHEVs includes EREVs. Extended-range EVs are a subset of plug-in hybrid electric vehicles that have both an internal combustion engine (ICE) and a plug-in rechargeable battery. Throughout this report, EREVs refer to plug-in [series](#) hybrid powertrain configurations where the ICE only operates without direct mechanical connection to the wheels to recharge the battery through an electric generator. Other plug-in hybrid configurations (such as parallel and series-parallel [configurations](#)) fall under the "standard" PHEV category. EREVs generally feature larger battery and longer electric range than standard PHEVs.

2024. The acceleration of PHEV sales in China led the share of electric car sales that are battery electric to fall from 80% in 2020 to below 60% in 2024, though in absolute terms battery electric car sales increased sevenfold over the same period, demonstrating their continued appeal to new customers.

In Europe, electric car sales stagnated in 2024 as policy support waned in major car markets

About one in five new cars sold on the European market was electric in 2024, maintaining the sales share of the previous year. The electric sales share increased in 2024 in 14 out of 27 EU member states, while it either stalled or decreased in the rest, including in several larger markets, such as Germany and France, largely as a result of subsidies being phased out or reduced. In [Germany](#), subsidies ceased at the end of 2023, while [France](#) has progressively reduced its subsidy over the years. At the start of 2024, France limited the amount of environmental bonus available to higher-income car buyers and reduced the number of vehicles eligible for the subsidy.

Besides subsidies, the policy design of the European Union CO₂ standards may also have held back further growth of the electric car market in 2024. As new targets come into effect every 5 years, car makers had [no incentive](#) to push sales of electric cars further in 2024 (in anticipation of strengthened targets in 2025). This is in contrast to markets such as the United Kingdom, where annually increasing targets move original equipment manufacturers (OEMs) towards electrification each year. In March 2025, the European Commission published its [Industrial Action Plan](#) for the European automotive sector, in which it proposed to amend the CO₂ emission performance standards for cars and vans, granting them additional flexibilities by averaging their performance over a 3-year period.

In the United Kingdom – the second-largest car market in Europe – electric car sales reached a share of nearly 30%, up from 24% in 2023. The year 2024 was the first under the [Vehicle Emissions Trading Scheme](#), which required 22% of all new registrations to be BEV or fuel cell electric vehicle (FCEV). When accounting for the scheme's flexibilities, which allows OEMs to borrow credits from future years, they were able to [comply](#) with a battery electric car sales share of close to 20%. Norway reached near-total electrification of sales, with [88%](#) of car sales being battery electric and just under 3% plug-in hybrid. As a result of the growing stock of electric cars, Norway's oil consumption for road in 2024 decreased by [12%](#) compared to 2021. From April 2025, [a tax increase](#) on conventional internal combustion engine (ICE) cars and PHEVs is expected to further increase the battery electric share towards meeting the Norwegian government's 2025 goal of 100% zero-emissions car sales. In Denmark, the electric sales share increased by 10 percentage points to reach 56%, with nearly 100 000 electric cars sold.

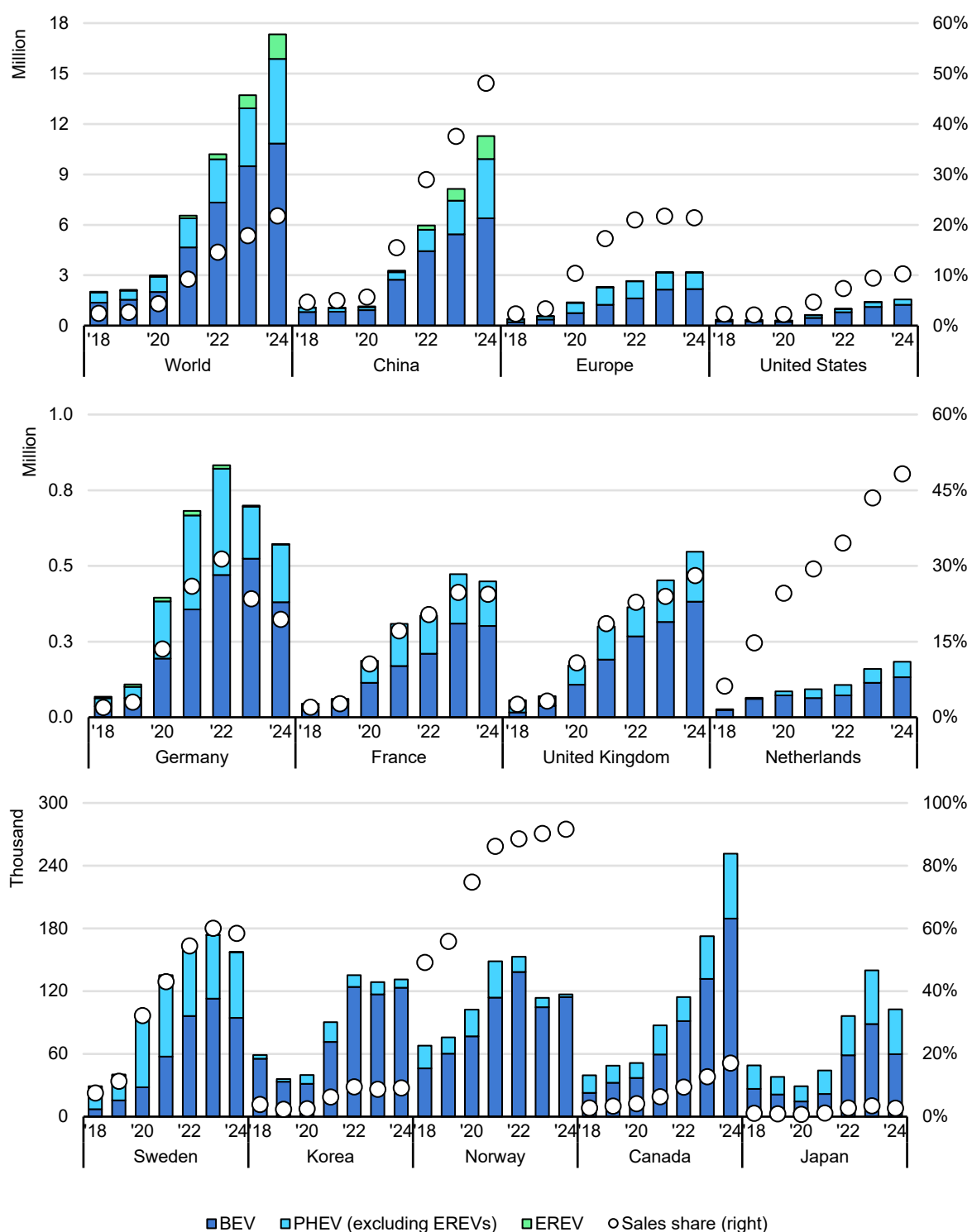
The market share of electric cars continued to expand in the United States

In the United States, electric car sales increased to 1.6 million in 2024, with the sales share growing to more than 10%. However, growth in electric car sales slowed down significantly in 2024, increasing by just 10% compared to 40% in 2023. In spite of this, electric car sales did boost the overall car market, as sales of conventional cars stagnated.

A total of 24 new electric car models were launched in 2024, increasing model availability by 15% compared to 2023, providing consumers with more choices and further increasing competition. While the Tesla Model Y and Model 3 have been the two best-selling models in the United States since 2020, the 110 new models that have entered the market since then have driven the market share of Tesla down from 60% in 2020 to 38% in 2024. Furthermore, 2024 was the first year in which Tesla saw a drop in sales in the United States, while other OEMs saw sales increase by 20% on aggregate.

A modification to the US Clean Vehicle Tax Credit at the start of 2024 enabled buyers to receive an instant discount (up to USD 7 500 for a new electric car and USD 4 000 for a used electric car) at the point of sale, which may have served to entice interested buyers. However, not all electric cars were eligible for the credit: in 2024 about 20 electric models (not accounting for different trim levels) out of a total 110 were eligible, which translated to over half of US electric car sales. The real share that benefitted from the tax credit may be even higher. In 2023, provisions were introduced on leased electric cars to [reclassify](#) them as commercial vehicles, thereby making them eligible for the tax credit without having to meet requirements on local manufacturing. As a result, by 2024, nearly [half](#) of all EVs sold were leased, more than double the share seen 3 years earlier. In addition to the federal tax credit, in 2024, [27 states offered additional incentives](#), rebates and exemptions promoting electric car adoption.

Electric car registrations and sales share in selected countries and regions, 2018-2024



IEA. CC BY 4.0.

Notes: BEV = battery electric vehicle; PHEV = plug-in hybrid vehicle, EREV = extended-range electric vehicle.

Sources: IEA analysis based on country submissions and data from [ACEA](#), [EAFO](#), [EV Volumes](#) and [Marklines](#).

Emerging markets saw a strong uptick in electric car sales

Electric car sales shares doubled in a number of emerging markets

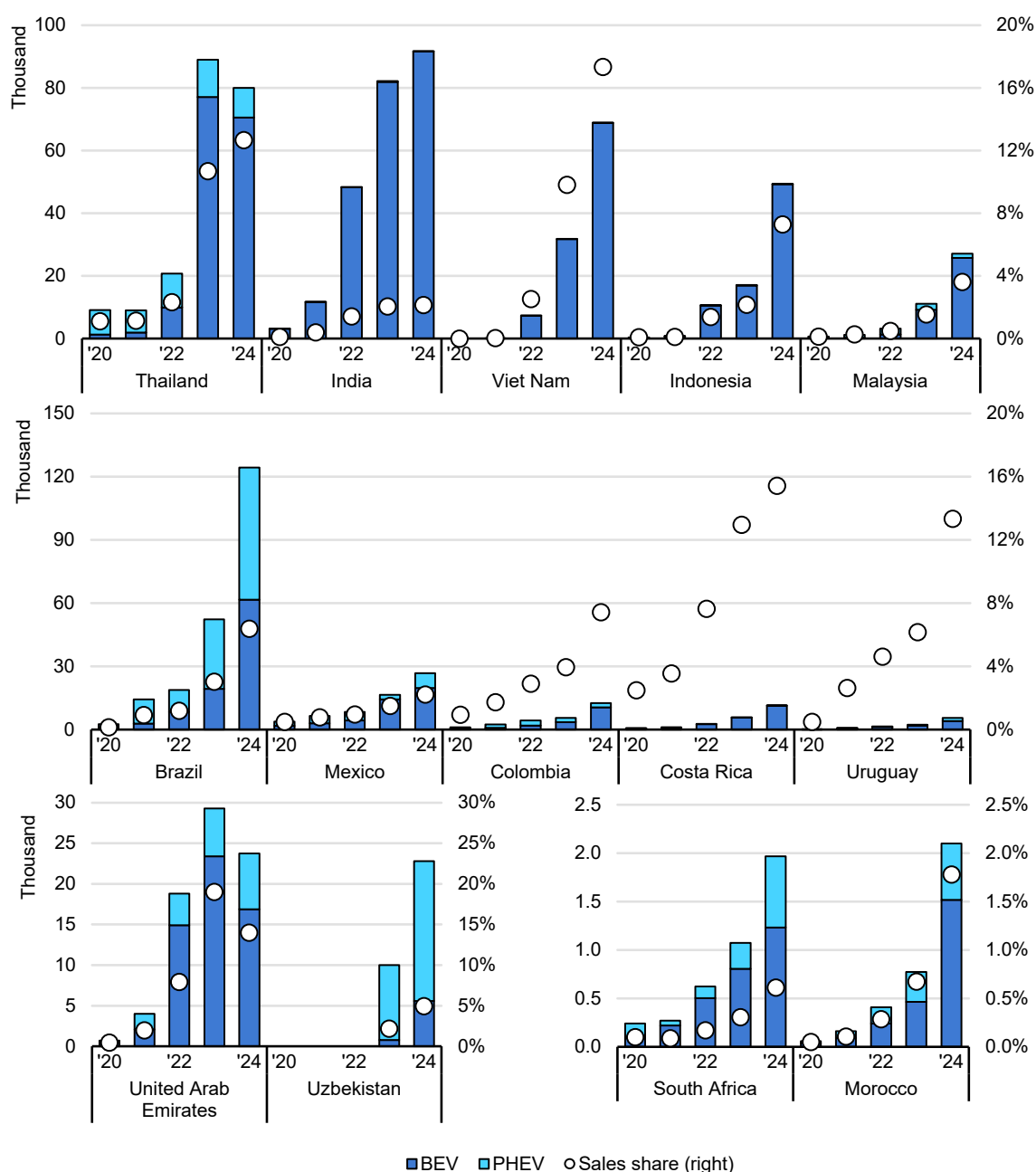
In emerging and developing economies in Asia, Latin America and Africa, electric car sales increased by over 60% year-on-year in 2024, and the sales share almost doubled from 2.5% to 4%. This rapid growth has been strengthened by policy incentives and the growing presence of relatively affordable electric cars from Chinese OEMs.

Emerging and developing economies in **Asia** (excluding China) saw a large increase in electric car sales, reaching almost 400 000 in 2024, up over 40% from 2023. However, in India, total electric car sales and their share of sales increased only slightly, approaching 100 000 (or 2%) in 2024. Thailand remained the largest EV market in Southeast Asia, despite a 10% drop in electric car sales. This decline was outweighed by an even steeper 26% drop in conventional car sales, largely due to [stricter lending criteria](#), meaning the electric sales share rose to 13% in 2024, up from 11% the previous year. Within the region, Indonesia and Viet Nam also stood out, respectively tripling and nearly doubling their sales numbers and reaching sales shares comparable to countries such as Spain or Canada. In many Southeast Asian countries, BEVs are the most popular electric car type, with over 90% of all electric car sales being fully electric.

In **Latin America**, sales volumes and penetration rates doubled in many countries, with electric cars reaching a market share of 4% in 2024. Brazil towered over other countries in the region with nearly 125 000 electric car sales, more than twice the number of 2023 sales, and the electric sales share doubled to 6.5%. Costa Rica, Uruguay and Colombia also achieved impressive sales shares of around 15%, 13% and 7.5%, respectively. These increases are in large part the result of government [incentives](#) such as tax exemptions, reduced registration fees, a relaxation of traffic restrictions for EVs, and relatively high fossil fuel prices.

In **Africa**, electric car sales more than doubled to reach nearly 11 000 in 2024. Sales shares remained low, at under 1%, though there was growth in several countries, such as Morocco and Egypt, where new electric car sales increased to more than 2 000.

Electric car registrations and sales shares in selected countries, 2020-2024



IEA. CC BY 4.0.

Notes: BEV = battery electric vehicle; PHEV = plug-in hybrid electric vehicle.

Sources: IEA analysis based on country submissions and data from ACEA, EAFO, EV Volumes, Marklines, [Asomove](#), [AleTech](#), [Andemos](#), OICA, [AFMA](#), [Gaikindo](#) and [AIVAM](#), sales of other macro regions can be found in the [Global EV Data Explorer](#).

Emerging electric car brands are seeing local and even international success

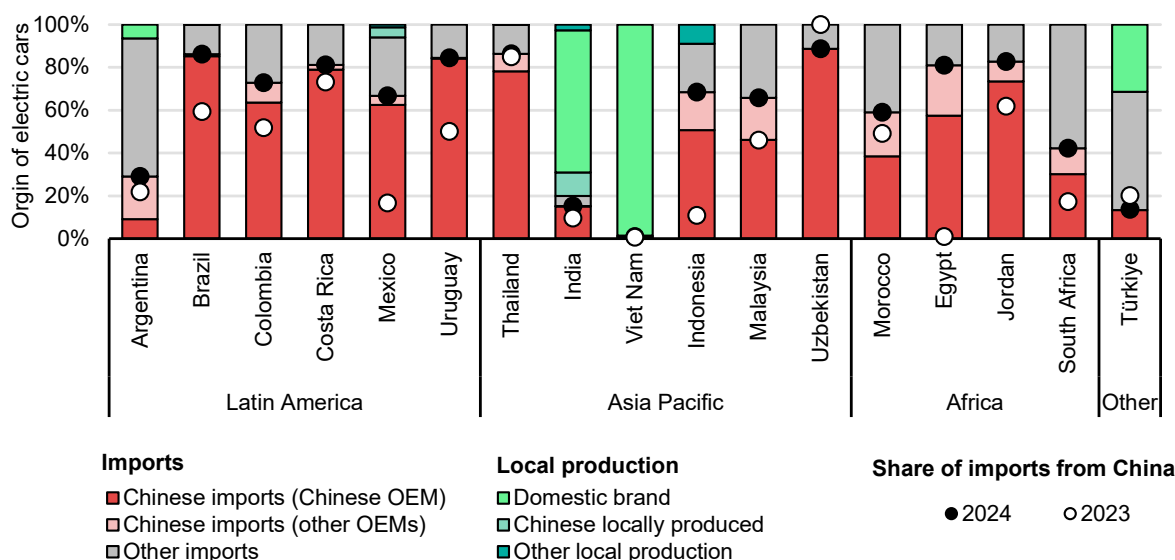
Recent years have seen the birth of several new electric car brands in emerging markets, such as VinFast in Viet Nam, Togg in Türkiye, and [Tito](#) in Argentina, helping to drive up sales. In **Viet Nam**, local brands underpinned the near doubling

of the electric sales share in 2024 to reach 17%. VinFast has also started to export to 11 different countries, with the largest share of exports going to Southeast Asian countries such as Indonesia (44%) and Malaysia (22%), but also to the United States (22%). The company has announced plans to [double](#) domestic production in Viet Nam, as well as expanding manufacturing into [India](#) and Indonesia.

Türkiye continues to deploy the domestically produced Togg electric cars, increasing its total production by 50% year-on-year. The increase in domestic production and in imports resulted in the country reaching an electric sales share of 10% in 2024. In July 2024, the Turkish government announced a [USD 5 billion](#) package to boost its annual production to 1 million electric cars per year.

India's electric car market is predominantly supplied by Indian OEM Tata. In addition, at the end of 2023, Indian steelmaker JSW signed a joint venture with SAIC Motor, a Chinese OEM, to work together on car production under the name MG Motor. In 2024, the joint venture produced about half of its electric cars sold in India domestically, while the other half were imported from China. At the end of 2024, [JSW](#) also announced the launch of its own EV brand, and is in talks with partners such as BYD and Geely regarding [collaboration](#) on technology transfer.

Origin of electric cars sold in selected markets, 2024, and share of total imports from China, 2023 and 2024



IEA. CC BY 4.0.

Notes: Chinese OEMs include BAIC, Geely-Volvo, GWM, GAC, BYD, Chery Automobile, JAC, Neta Auto, Seres Group, FAW, Changan, Dongfeng, Jiangling Motors, SAIC, Leap Motor, Xiaopeng, Aiyangs Automobile. The domestic brands are Tata Motors, Mahindra & Mahindra (India); Togg Inc. (Türkiye); VinFast (Viet Nam) and Tito (Argentina).

Sources: IEA analysis based on data from EV Volumes.

Overseas expansion plans of Chinese producers, prompted by changes to import tariffs, drove sales in emerging markets

Another notable trend from 2024 was the numerous manufacturing announcements from Chinese OEMs in countries such as Brazil, Thailand, Indonesia and Malaysia, where temporary exemptions from import tariffs for electric cars are coming to an end. In 2024, electric car sales in **Brazil** more than doubled, with over 85% of new electric cars coming from China. A key factor behind this impressive growth was electric cars being exempt from import duties of 35%, though this exemption started to be gradually removed in 2024 and is set to end by the middle of 2026. By then, automakers [BYD](#) and [GWM](#) will have begun producing models in Brazil, with a range tailored to the Brazilian market, such as BYD's [Song Pro](#), which will be flex-fuel compatible.

In **Thailand**, Chinese imports play a key role in electrification, accounting for 85% of electric car sales. However, as in Brazil, this share is expected to decline due to changes introduced in the [EV 3.5 program](#), under which import taxes on electric cars will be reintroduced at the end of 2025 and existing subsidies of up to THB 100 000 (Thai baht) (USD 2 800) per imported car will be gradually phased out. To continue to qualify for the [exemption and subsidy](#), OEMs must commit to producing at least two battery electric vehicles (BEVs) domestically by the end of 2026 for every imported BEV sold in 2024 and 2025.

In **Indonesia**, electric car sales tripled in 2024 while the conventional market contracted by 20%, leading to an electric sales share of over 7%. The government has supported adoption by reducing the VAT rate on electric cars from 11% to 1% since 2023. However, what has potentially had an even [greater impact](#) on reducing the cost of electric cars for Indonesian buyers – and thus stimulating EV sales – has been the government waiving [import taxes](#) for EVs from car makers that invest in local electric car manufacturing from the start of 2024. Chinese car manufactures such as [BYD](#) and [GAC Aion](#) profited from this reduction, as did European OEM [Stellantis](#). As a result, Chinese electric car imports increased 18-fold to 34 000 by the end of 2024.

In **Malaysia**, electric car sales more than doubled in 2024 and the sales share increased from less than 2% in 2023 to nearly 4% in 2024. Much of this success is due to imported electric cars being [exempt from import taxes and excise duties](#) until the end of 2025. Ahead of the tax exemption ending, Chinese car producer Geely, together with Proton (a Malaysian automotive company), will start domestically [producing](#) the e.MAS 7 by the end of 2025, backed by [an investment](#) from Geely. Malaysian OEM [Perodua](#) will also start producing its first electric car at the end of 2025, with a price of RM 80 000 (Malaysia ringgit) (USD 18 000).

More affordable electric cars and local policies have reshaped the EV landscape in more emerging markets

Policy measures such as [tax incentives](#) for electric cars contributed to the doubling of electric car sales seen in **Colombia** and **Costa Rica** in 2024. Colombia also proposed to increase [import tariffs](#) on all hybrid vehicles at the end of 2024, in a push to shift more sales towards BEVs. More than 70% of all electric cars sold in these two countries were imported from China, with Chinese imports to Colombia more than tripling in 2024, while imports from other countries grew by just 30%. In **Uzbekistan**, electric car sales shares doubled and the [average price of an imported electric car](#) fell almost threefold between 2023 and 2024.

In **Africa**, local policies and changing trade regimes are reshaping the electric car market. Developments such as the ban on imports of petrol and diesel cars introduced in **Ethiopia** at the start of 2024 have resulted in a reported deployment of [100 000](#) electric vehicles.³ While model availability appears to be reasonable in the country, there are [reports](#) that garages struggle to source components for repairs, and that deployment of chargers outside of the capital has not kept pace with electric car sales. In **Morocco** and **Egypt**, efforts by car manufacturers to expand their production lines for batteries and/or electric cars in order to facilitate exports to the European Union have also pushed up domestic deployment, with the sales share reaching just under 2%. **Nigeria** is now looking into strengthening its EV manufacturing capacity with support from Morocco, and in 2024, signed the [Zero Emission Vehicles Declaration](#) to work towards all new sales of cars and vans being zero emission by 2040.

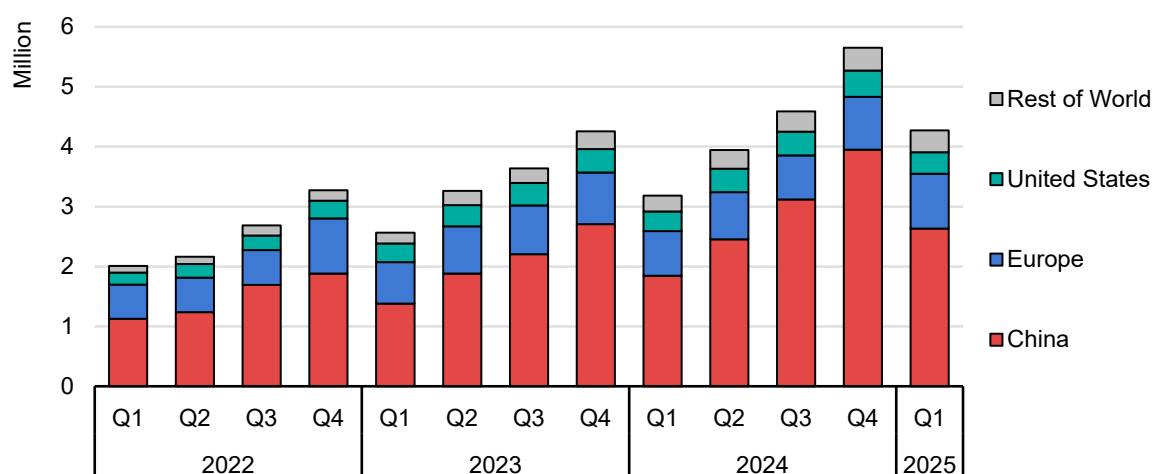
First quarter sales hint at strong sales for 2025, despite many uncertainties

Electric car sales increased 35% in the first quarter of 2025 compared to the same period in 2024

More than 4 million electric cars were sold in the first quarter of 2025 as sales grew by 35% compared to the first quarter of 2024, which was higher than the growth rate observed in the first quarters of the previous 2 years. Over 1 million more electric cars were sold in the first three months of 2025 compared to the same period in 2024 and about 60% of these were sold in China.

³ Data on electric car deployment in Ethiopia is difficult to obtain, and sources are not well aligned. The Ministry of Transport of Ethiopia reports a stock of 100 000 electric vehicles but does not define how many of these are cars. In contrast, EV Volumes reports about 1 300 electric car sales in the past 5 years.

Quarterly electric car sales, 2022-Q1 2025



IEA. CC BY 4.0.

Sources: IEA analysis based on data from EV Volumes.

Between January and March 2025, China averaged monthly sales of around 875 000 electric cars, with total sales of more than 2.5 million. While the electric sales share was less than 45% in January, it reached more than 50% in both February and March. Across the full first quarter, the share of electric car sales was similar to that of 2024.

In Europe, electric car sales reached more than 900 000 in the first quarter of 2025, 625 000 of which were sold in the European Union. The electric sales share has also been increasing, averaging one in four cars sold in Europe in the first three months of 2025. In the United Kingdom, electric cars represented 30% of cars sold in the first quarter, while in the European Union the share was less than 25%.

More than 360 000 electric cars were sold in the United States in the first three months of 2025, around 10% more than during the same period the previous year. Total car sales grew at a similar rate, meaning the electric car sales share in the United States remained around 10%, the average across 2024.

Several emerging electric car markets continued to see strong growth in the first quarter of 2025. In Brazil, for example, electric car sales exceeded 30 000 across the first three months of 2025, 40% more than during the same period in 2024. Sales in India grew 45% year-on-year, nearing 35 000 electric car sales for the first quarter of 2025. Electric car sales in Viet Nam also approached 35 000 for the first quarter – nearly four times as many as were sold during the first quarter of 2024 – leading sales across Southeast Asia to double to over 100 000 between January and March 2025.

Electric car sales are expected to exceed 20 million in 2025, representing one-quarter of total car sales

For the full year 2025, electric car sales are expected to increase by 25% globally, which is similar to the growth rate from the 2024. As a result, electric car sales top 20 million worldwide. While sales volumes may be impacted by economic and policy [uncertainties](#), more than one in four cars sold in 2025 is expected to be electric.

Building on strong sales in the first quarter of 2025, China is expected to sell over 14 million electric cars across the full year – more than were sold globally in 2023. Sales of both electric and conventional cars are supported by the extension of the trade-in scheme for older vehicles. The sales share of electric cars in China is expected to reach around 60% in 2025.

The European Commission announced in March 2025 that OEMs would be given flexibilities in meeting the 2025 CO₂ targets for cars and vans. On this basis, OEMs now only need to achieve the 15% emissions reduction target (compared to the 2021 baseline) on average over 2025-2027. This announcement was made in response to [auto industry claims](#) that the 2025 target would be unachievable, and would thus incur significant financial penalties that would damage the already struggling European auto industry. As yet, it is unclear how different OEMs will approach meeting the 3-year average target, in terms of the degree of under-performance that could be accepted in 2025 but would then need to be compensated for in the following 2 years.

The new phase of the CO₂ targets in the European Union is expected to drive up electric car sales, although the new flexibilities mean OEMs have less of an incentive to bring lower-priced electric cars to market this year. Given that a purchase premium for electric cars [persists](#) in Europe, the phase-out of EV purchase subsidies in some European countries could place downward pressure on sales, but there are signs to the contrary. At the end of 2024, the electric car purchase subsidy in the [Netherlands](#) came to an end, but electric car sales in the country in the first quarter of 2025 were about 10% higher than during the same period of 2024. In Italy, direct purchase subsidies for electric cars are no longer being renewed after 2024, though the government does aim to support the domestic production of EVs. In the first three months of 2025, electric car sales in Italy were up almost 50%.

Over the course of the year, electric car sales are expected to reach around 4 million across Europe. In line with this, the share of electric cars sales would increase several percentage points compared to 2024, to around 25%. The increase in electric car sales across Europe is boosted by the UK Vehicle Emissions Trading Scheme, which sets the target of 28% battery electric car sales in 2025.

In the United States, [Executive Order 14154](#) directed the government to reconsider market interventions that favour EVs. Legislation has since been proposed that would end the Clean Vehicle Tax Credit for both cars and light commercial vehicles. In 2025, this proposal may result in consumers that have been considering buying an electric car rushing to do so before the tax credit is removed. A dampening effect on EV sales in the United States is expected only once the tax credit is repealed, and the timeline for that is uncertain. In addition, [tariffs](#) for conventional and electric cars have also been announced, which may result in lower car sales. For the full year, electric car sales in the United States are expected to grow almost 10% in 2025, with a slight increase in the electric car sales share.

Across the rest of the world, electric car sales are expected to grow by over 30% to around 1.8 million. This would mean electric cars account for 6% of all car sales outside of the three major EV markets in 2025, up from 5% in 2024. The strong growth seen in Southeast Asia and Brazil is expected to continue, reaching sales of more than 0.5 million combined in 2025.

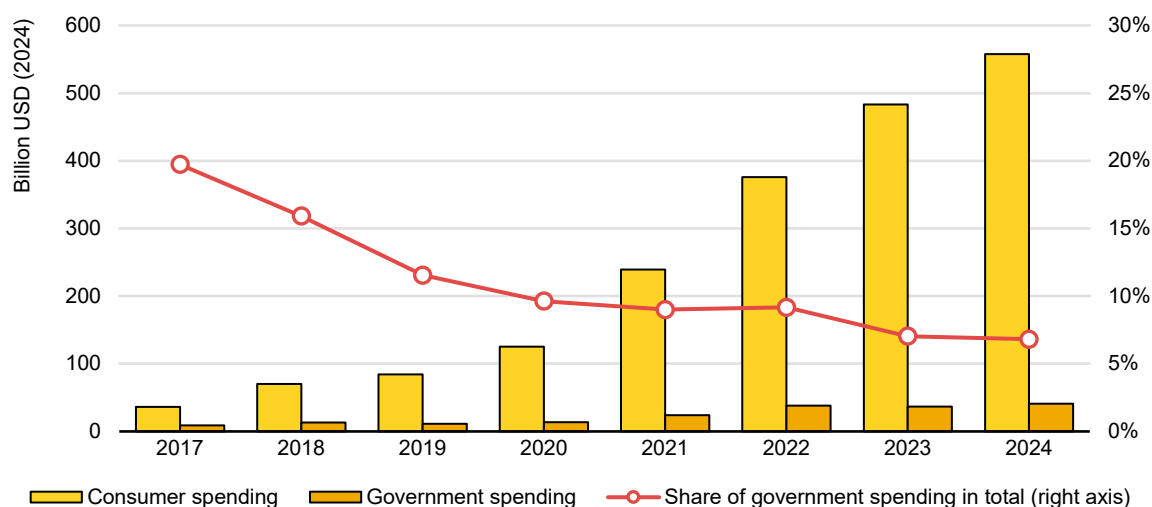
Government spending on electric cars

Subsidy phase-outs are leading to ever lower shares of government spending in the EV market

As electric car sales have grown over the past decade, government spending per vehicle, in the form of purchase subsidies and tax incentives, has steadily declined – a trend that accelerated in 2022 as subsidies were phased out.

In 2024, government spending accounted for less than 7% of total spending on electric cars globally, compared to 20% in 2017. In absolute terms, annual government spending on electric cars has hovered around the USD 38 billion mark since 2022. At the same time, total consumer spending on electric cars globally has grown continuously to reach USD 560 billion in 2024.

Global consumer and government spending on electric cars, 2017-2024



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Notes: Government spending is the sum of direct central government spending through purchase incentives and foregone revenue due to purchase and registration taxes waived specifically for new electric cars. Only central government purchase support policies for electric cars are taken into account. Spending on charging is not included. Consumer spending is the total expenditure based on model price, minus government incentives. Values and trends may change slightly relative to previous publications following methodology improvements and better coverage of government support schemes.

Sources: IEA analysis based on EV Volumes and national policy documents.

China has had the largest absolute public spending since 2020, despite its subsidy scheme coming to an end in December 2022 after 12 years. Other incentives have remained in place – notably, electric cars being exempt from the 10% purchase tax that applies to other vehicles. In 2024, China introduced a trade-in subsidy with a higher premium for the purchase of EVs. The additional expenditure to support EV purchases is estimated to be around USD 2.7 billion for 2024.⁴ Nevertheless, Chinese government expenditure per vehicle dropped 25% between 2022 and 2024. China has the highest share of subsidies going to PHEVs, at nearly 45%, as a result of the 10% purchase tax exemption being applied to PHEVs, which are on average more expensive than BEVs. In contrast, government incentives for PHEVs have almost disappeared from other key markets.

In Europe, subsidy schemes were significantly reshaped in 2023 and 2024. The most abrupt change was in Germany, where the EUR 4 500 per-vehicle subsidy was reduced to zero as of December 2023 – which was followed by a drop in EV market share of 4 percentage points in 2024. As a result, the German government

⁴ The [original](#) funding allocation for the car trade-in subsidies in 2024 was close to CNY 11.2 billion (USD 1.5 billion), however, for the full year, [6.6 million](#) cars were traded in. The additional expenditure for electric car purchases is estimated by applying the CNY 5 000 subsidy differential to the [60%](#) of trade-ins that were for the purchase of an electric car.

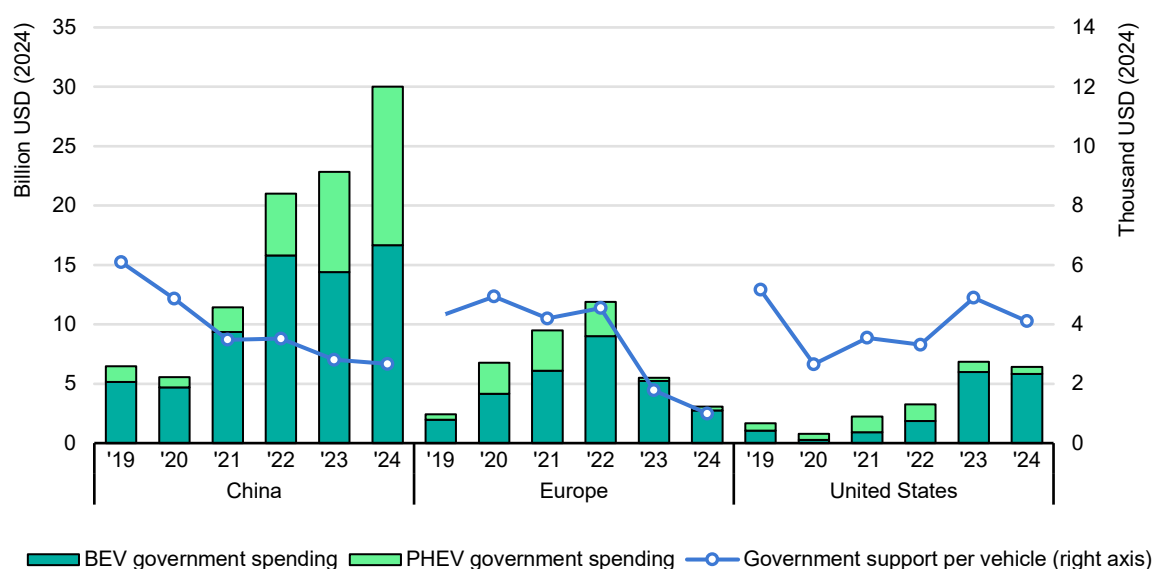
has introduced new [tax benefits](#) for companies buying EVs, effective from July 2024 through to 2028. Company cars account for about [two-thirds](#) of registrations in the country.

In the United Kingdom, all subsidies were removed at the end of 2022 but electric car sales have continued to increase, thanks in part to the [Vehicle Emissions Trading Schemes](#), which set targets for zero-emission car sales starting from 2024. In addition, the growth in EV sales was underpinned by a set of tax rebates for company cars, which represented about [60%](#) of total car registrations in 2024. When factoring in all [government tax incentives](#), an ICE car would incur more than [10 times](#) the company car tax of an equivalent BEV, compared to Germany where this ratio is around [4](#).

Similarly, Sweden ended its subsidy programme offering up to SEK 60 000 (Swedish kronor) (USD 5 800) per electric car in November 2022. Absolute volumes of electric car sales in the country decreased 10% in 2024, though the country's EV sales share fell only very slightly due to a drop in overall car sales. In several other European countries, subsidies were either reduced in absolute terms, or their conditions have become more stringent, with a faster roll-back of subsidies for PHEVs. As a result, the average subsidy per vehicle in Europe has fallen rapidly, from more than USD 4 500 per vehicle in 2022 to around USD 1 000 in 2024. The phase-down or interruption of subsidy schemes in Europe has had differing consequences on EV market shares at the national level, but the rapid decrease in subsidies has not been followed by an equivalent decrease in EV market share, providing an encouraging sign of market resilience.

In the United States, the introduction of the Inflation Reduction Act expanded the number of vehicles eligible for tax credits in 2023. However, the list of vehicles eligible for the USD 7 500 tax credit has since changed repeatedly as Internal Revenue Service guidelines have been implemented. The average tax credit in 2024 was approximately USD 4 000 per vehicle, since not all electric cars were eligible and some were eligible for only USD 3 750. In both 2023 and 2024, the United States had the highest per-vehicle subsidy of the three major EV markets.

Government spending on electric cars by region and powertrain, 2019-2024



IEA. CC BY 4.0.

Notes: BEV = battery electric vehicle; PHEV = plug-in hybrid electric vehicle. Government spending is the sum of direct central government spending through purchase incentives and foregone revenue due to purchase and registration taxes waived specifically for new electric cars. Only central government purchase support policies for electric cars are taken into account. Spending on charging is not included. Consumer spending is the total expenditure based on model price, minus government incentives. Excludes incentives for company cars. Values and trends may change slightly relative to previous publications following methodology improvements and better coverage of government support schemes.

Sources: IEA analysis based on EV Volumes and national policy documents.

7. Electric vehicle batteries

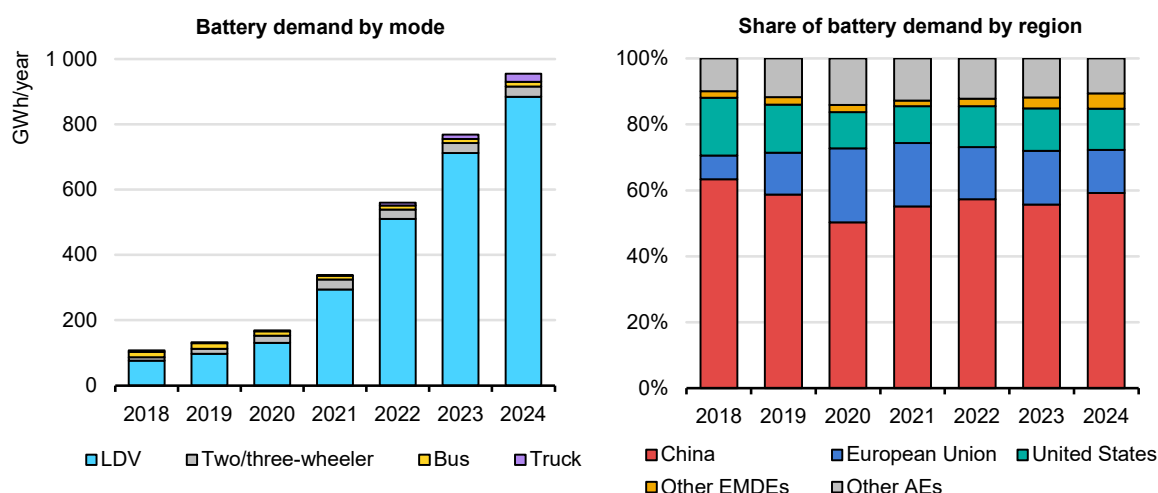
Trends in battery demand

Global battery demand for the energy sector hit the 1 TWh milestone in 2024

Electric cars remain the main driver of battery demand, but demand for trucks nearly doubled

Battery demand in the energy sector, for both EV batteries and storage applications, reached the historical milestone of 1 TWh in 2024. Demand for one average week alone in 2024 exceeded the total demand for an entire year just a decade earlier. Demand was largely driven by growth in EV sales, as demand for EV batteries grew to over 950 GWh – 25% more than in 2023. Electric cars remain the principal factor behind EV battery demand, accounting for over 85%. Compared to 2023, the sector whose demand grew the most was electric trucks, growing over 75% in 2024 to reach nearly 3% of global EV battery demand. Electric truck battery demand was driven by growth in China, but demand also ramped up in Europe (about 25%), which accounted for about 10% of the global total.

Electric vehicle battery demand by mode and region, 2018-2024



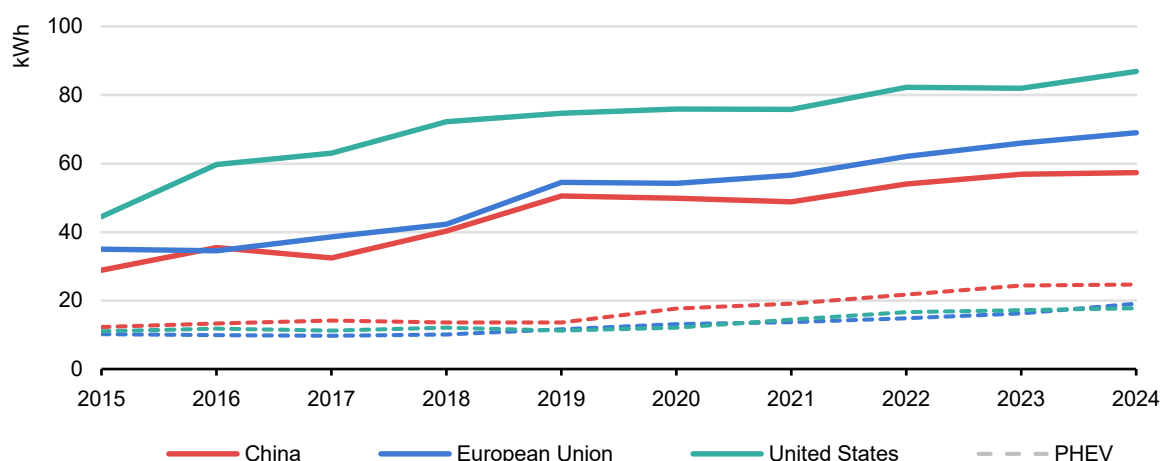
IEA. CC BY 4.0.

Notes: LDV = light-duty vehicle, including cars and vans; EMDEs = Emerging markets and developing economies; AEs = Advanced economies. Battery demand is defined here as the volume-weighted average battery size multiplied by vehicle sales by mode and region. It reflects the batteries installed in vehicles sold in each region, and not the battery demand for vehicles manufactured in each region. Electric vehicle and battery stockpiling are excluded from the analysis.

Source: IEA analysis based on [EV Volumes](#).

In 2024, EV battery demand grew by over 30% in China, and by 20% in the United States, in stark contrast with the European Union, where demand stalled. Battery demand in the United States nearly matched that of the European Union in 2024, in part as a result of its approximately 25% larger battery size per EV. Emerging markets and developing economies other than China continued to represent only a small share of global battery demand, reaching nearly 5% in 2024. Nevertheless, their share has doubled since 2022, underpinned by sustained growth in Southeast Asia, India and Brazil.

Average electric car battery pack size by region and powertrain, 2015-2024



IEA. CC BY 4.0.

Notes: PHEV = plug-in hybrid electric vehicle. Solid lines represent the sales-weighted average battery pack size for battery electric cars, while dashed lines indicate the same for plug-in hybrid electric cars.

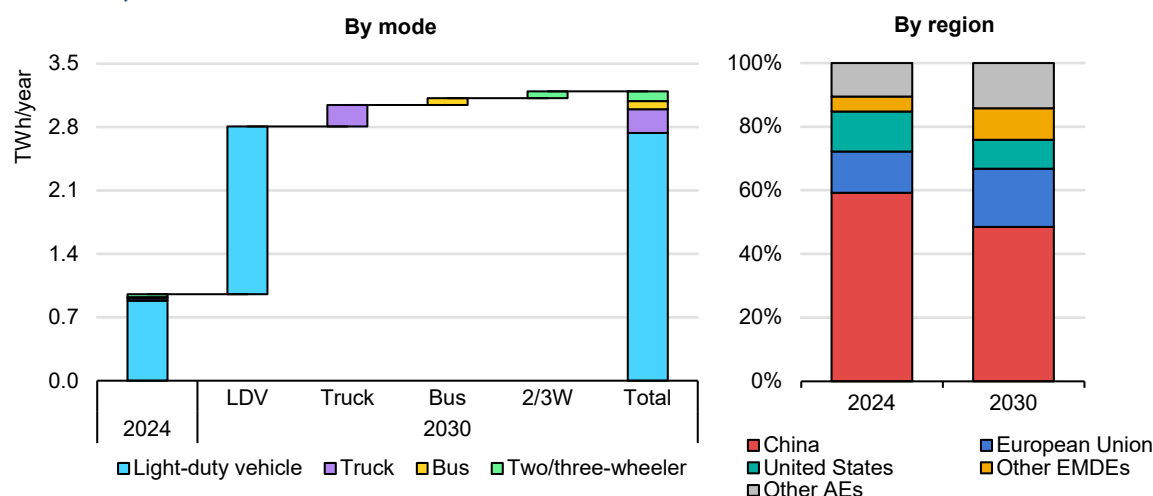
Source: IEA analysis based on [EV Volumes](#).

Outlook for battery demand

Electric vehicle battery demand jumps more than threefold by 2030

EV battery demand continues to grow, and is expected to reach more than 3 TWh in 2030 in the STEPS, up from about 1 TWh in 2024. While electric cars will remain the primary driver of battery demand, other modes are set to gain market share. Notably, the contribution of electric trucks to EV battery demand triples by 2030 to reach more than 8%, up from nearly 3% in 2024.

Battery demand for electric vehicles by mode and region in the Stated Policies Scenario, 2024-2030



IEA. CC BY 4.0.

Notes: LDV = light-duty vehicle, including cars and vans; 2/3W = two/three-wheeler; EMDEs = emerging markets and developing economies; AEs = advanced economies. Battery demand is defined here as the volume-weighted average battery size multiplied by vehicle sales by mode and region. It reflects the batteries installed in vehicles sold in each region, and not the battery demand for vehicles manufactured in each region. Electric vehicle and battery stockpiling are excluded from the analysis.

Battery demand is also set to become more geographically diverse. In the STEPS, emerging markets and developing economies other than China are expected to double their share of EV battery demand, from nearly 5% in 2024 to 10% in 2030. The share of global demand in the European Union and other advanced economies, such as the United Kingdom, Canada, Japan and Korea, is also projected to grow, while the share of the United States is projected to decline from about 13% in 2024 to less than 10% by 2030. Meanwhile, China's share of global battery demand declines from 60% in 2024 to just under 50% by 2030, although it remains by far the single largest source of demand.

Battery industry trends

Chinese manufacturers are increasing their competitive advantage

Low critical mineral prices and intense competition drove down battery prices in 2024, but China's price advantage is widening

Prices for lithium-ion battery packs fell 20% in 2024 – the largest drop since 2017 – as a result of low critical mineral prices and battery margins being [squeezed](#) through competition, predominantly in China. Lithium prices, in particular, dropped

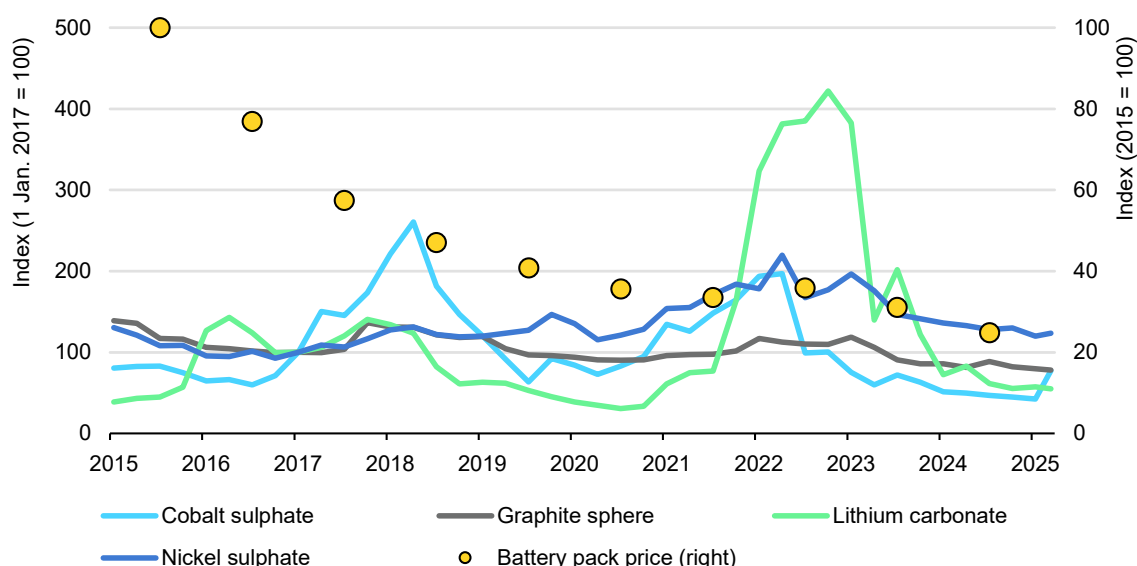
nearly 20% in 2024, reaching similar prices to those at the end of 2015, despite lithium demand in 2024 being about six times bigger than in 2015.

Low critical mineral prices are primarily driven by supply surplus, which is making it [difficult](#) for some mining companies to compete, thus increasing the level of supply chain concentration among established players. This surplus is [expected](#) to persist over the next few years, but low prices could discourage future investments and might cause supply [shortages](#) for [lithium](#) and [nickel](#) by 2030. In addition, the high [geographical](#) and [ownership](#) concentration of their supply chains may cause market distortions, increasing market risk.

An undersupply of lithium would push up prices, to the benefit of the mining sector but to the detriment of battery and EV makers, as well as final consumers. The recycling sector, which could help curb cost increases, would also benefit from higher mineral prices. However, due to feedstock limitations, it will take about a decade before recycling has a significant impact on reducing primary mineral demand (see the Box on [battery recycling](#) below).

[Technology innovation](#), particularly related to [sodium-ion batteries](#) or [direct lithium extraction](#), could be instrumental in reducing the risk of lithium undersupply and its potential impact, and avoiding price spikes similar to those seen in 2022. Additionally, vertical integration through [upstream investments](#) can help battery suppliers to lower production costs while safeguarding against the risk of volatile critical mineral prices.

Price of selected battery metals (left) and lithium-ion battery packs (right), 2015-2025



IEA. CC BY 4.0.

Notes: "Battery pack price" refers to the volume-weighted average pack price of lithium-ion batteries across the electric vehicles and battery storage sectors. 2025 refers to data up to the end of March 2025.

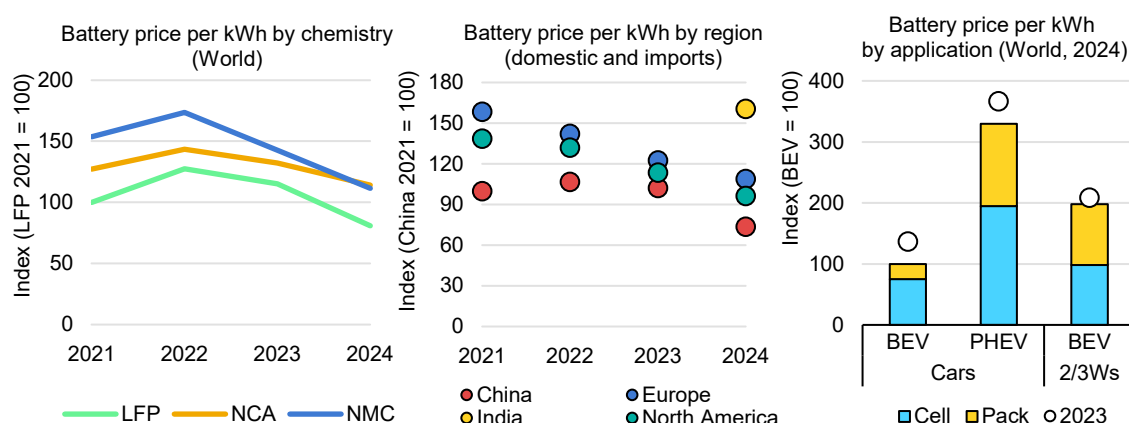
Sources: IEA analysis based on data from [Bloomberg](#) and [Bloomberg New Energy Finance](#) Lithium-Ion Price Survey (2024).

Battery pack prices fell in all markets, but the extent of the drop varied significantly, with the fastest declines seen in China, where prices fell nearly 30% in 2024, compared to 10-15% in Europe and the United States. This widened the gap between battery prices in China and the rest of the world, increasing the competitive advantage of Chinese EV and battery producers. The faster pace of battery [cost reduction](#) and [innovation](#) in China has been enabled by fierce competition that has driven [down](#) profit margins for most producers (though [not all](#)), at the same time as driving up manufacturing [efficiency and yields](#), as well as access to a large skilled workforce, and battery supply chain integration.

Battery chemistry also plays an important role, with lithium iron phosphate (LFP) batteries – the [main](#) battery chemistry used in China – being almost 30% [cheaper](#) per kilowatt-hour (kWh) than lithium nickel cobalt manganese oxide (NMC) batteries, which remain the most widely used batteries in the United States and Europe. NMC batteries still provide an energy density advantage, though the gap has [narrowed](#) in recent years. The energy density of LFP battery packs is about one-fifth lower by mass (Wh/kg) and about one-third lower by volume (Wh/L) than that of NMC packs. This is, however, partially offset by LFP's ability to reach [100%](#) state of charge when required without significant degradation, whereas NMC batteries are typically limited to [80%](#) to preserve long-term performance.

The higher energy density of NMC batteries remains an advantage for applications requiring longer ranges or operation in cold climates, where LFP technology is typically [less effective](#). However, LFP batteries have now reached a performance level sufficient for most EV applications, making their lower cost a key advantage for automakers aiming to mass markets.

Average battery pack price per watt-hour index by selected battery chemistry, region and mode, 2021-2024



IEA. CC BY 4.0.

Notes: LFP = lithium iron phosphate; NCA = lithium nickel cobalt aluminium oxide; NMC = lithium nickel manganese cobalt oxide; BEV = battery electric vehicles; PHEV = plug-in hybrid electric vehicle; 2/3Ws = two- and three-wheelers. Battery price refers to price per kWh for battery packs. Battery prices by region refer to the average battery price in a given region, including locally produced batteries and imports across EVs and battery storage applications. Data for India is not available between 2021 and 2023. Battery price by application refers to an average-sized vehicle.

Source: IEA analysis based on data from [Bloomberg New Energy Finance](#).

The battery price per kWh is also heavily dependent on the targeted application, with BEVs enabling the lowest costs. In 2024, battery pack prices per kWh for plug-in hybrid electric cars were more than three times those for battery electric cars, because of their smaller size and greater power requirements. In 2024, the average price of a 20 kWh PHEV battery pack – roughly the global sales-weighted average for standard plug-in hybrids – was about the same as a 65 kWh BEV battery pack, the global sales-weighted average for battery electric cars.

Pack components such as the battery management system are common to BEV and PHEV battery packs, but given that PHEV packs are smaller, the price of such components is spread across fewer battery cells, increasing the price per kWh. PHEV battery packs may also require more complex designs to accommodate the internal combustion engine, which increases their production cost. Additionally, because of the smaller pack size, each battery cell in the PHEV pack has higher power requirements to guarantee minimum acceleration standards while driving in full electric mode. This demands different battery designs, such as thinner electrodes, to improve battery power at the expense of its energy density,¹ further increasing the cost per kWh compared to BEVs. Smaller battery sizes also increase prices per kWh for 2/3W batteries.

Battery specifics, however, only determine prices for high volume. For instance, in China, electric truck battery prices per kWh are slightly lower than for battery electric cars, thanks to their larger size and therefore the reduced contribution of the battery pack cost. Electric truck markets in other countries are far less mature, and their battery price per kWh remains significantly higher – in 2024, prices were more than double those in China.

Lithium iron phosphate batteries continue to gain market share, and with them so do Chinese manufacturers

In 2024, LFP batteries made up nearly half of the global EV battery market, underpinning the efforts of manufacturers to [lower](#) EV prices and production costs in order to maintain or gain market share in an increasingly competitive market. China leads on the uptake of LFP batteries, which met nearly three-quarters of its domestic battery demand in 2024, and their share shows no sign of slowing down, reaching 80% of batteries sold in [November](#) and [December](#) 2024.

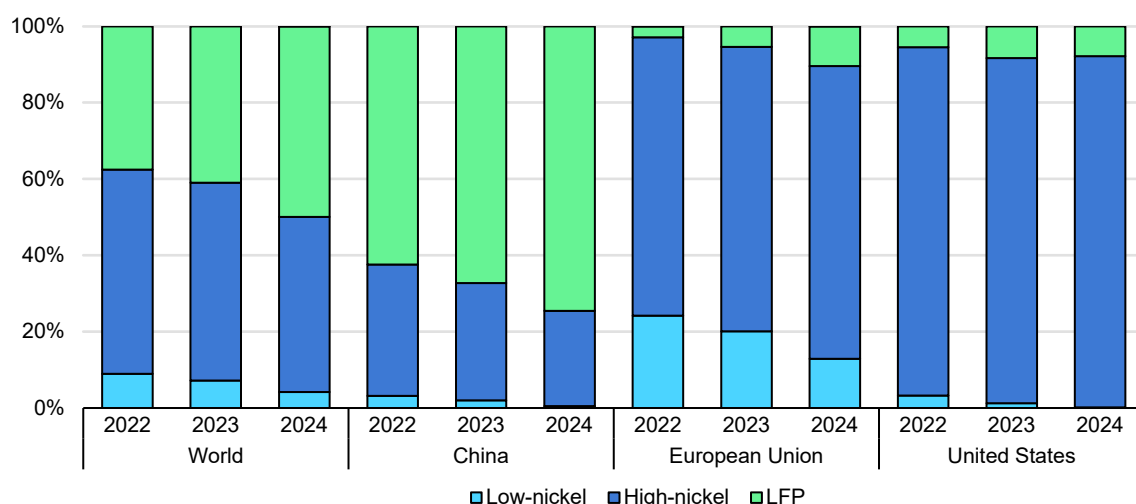
In the United States, the share of LFP batteries used in EVs slightly contracted in 2024, remaining below 10%, which may be a result of [tariffs](#) on Chinese batteries. In contrast, in the European Union, LFP battery adoption grew by about 90% for

¹ “Energy density” is used here as a general term referring to the amount of energy stored per unit of mass or volume. It can be divided into two specific metrics: specific energy (Wh/kg) and energy density (Wh/L). Specific energy influences the additional weight the battery adds to the vehicle, thereby affecting its efficiency. Energy density determines the amount of energy that can be stored in a battery pack of a given size, typically defined by the vehicle's chassis dimensions and shape.

the second consecutive year to reach more than 10% of the EU EV market. Notably, nearly all the LFP batteries for electric cars sold in Europe or the United States were produced in China, which today has a de facto [monopoly](#) on this type of battery. Market penetration of LFP batteries is moving even faster in other markets. In Southeast Asia, Brazil and India, the share of electric car batteries using LFP reached more than 50% in 2024. In Southeast Asia and Brazil, LFP uptake is led by imports from China, mostly by BYD, whereas in India it is driven by cars produced domestically, led by Tata Motors.

In the United States, the vast majority (three-quarters) of EVs equipped with LFP batteries are produced domestically, whereas in the European Union nearly two-thirds are imported from China. In both regions, LFP batteries are primarily sourced from China, with Tesla playing a central role – accounting for 85% of LFP battery-powered EVs produced in the United States and almost half of those sold in the European Union. Of Tesla's LFP EVs sold in the European Union, over half are produced in Germany, with the remainder imported from China.

Share of electric vehicle battery sales by chemistry and region, 2022-2024



IEA. CC BY 4.0.

Notes: LFP = lithium iron phosphate. Low-nickel includes lithium nickel manganese cobalt oxide (NMC) 333, NMC442, and NMC532. High-nickel includes NMC622, NMC721, NMC811, lithium nickel cobalt aluminium oxide (NCA), and lithium nickel manganese cobalt aluminium oxide (NMCA). LFP also includes lithium iron manganese phosphate (LFMP). Battery chemistry sales share is based on the battery capacity of EVs registered in the different regions. This calculation assumes that 95% of electric trucks, buses and light commercial vehicles sold in China use LFP, and that 70% of electric trucks and electric buses sold outside of China use high-nickel chemistries. Two/three-wheelers are excluded from the analysis. Battery sales reflect the batteries installed in vehicles sold in each region, and not the battery demand for vehicles manufactured in each region. Electric vehicle and battery stockpiling are excluded from the analysis.

Sources: IEA analysis based on data from [EV Volumes](#) and [China Automotive Battery Industry Innovation Alliance](#).

European OEMs are looking increasingly favourably on LFP batteries as a way of reducing production costs and are therefore interested in securing more supplies. Initiatives such as the recently announced [joint venture](#) between Stellantis and CATL for an LFP battery manufacturing plant of up to 50 GWh in Spain could

therefore help the European automotive industry to reduce costs and increase its competitiveness in the coming years. However, investments in LFP battery manufacturing in Europe might suffer as a result of the recent [proposition](#) from the Chinese government to limit exports of key battery technologies, including LFP cathode production and lithium processing. Overall, renewed interest in the LFP chemistry [exacerbates](#) the difficulties facing the European battery industry, which has historically focused on NMC batteries.

The year 2024 was a challenging one for the European battery industry, with both small and large producers suffering for different reasons. New players such as Northvolt, which filed for [bankruptcy](#) in the [United States](#) and [Sweden](#), faced serious difficulties in scaling up production, leading to insufficient manufacturing [yield](#) and high production costs. The case of Northvolt was further [aggravated](#) by limited experience, ambitious expansion plans, and excessively widened scope, leading to insufficient focus on scaling up high-quality battery cell production. Falling battery prices worldwide also weighed heavily on small producers, which have fewer resources and insufficient production volumes to withstand [lower profit margins](#), and are now turning to a more [cautious](#) growth path.

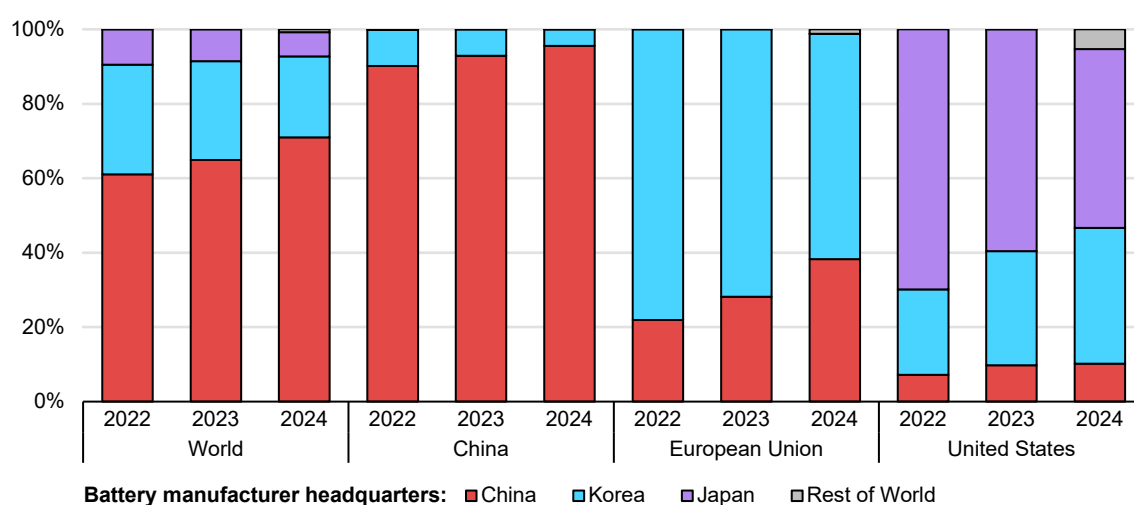
Korean companies, such as LG Energy Solution in Poland, are the largest battery producers in Europe, but their market share is also under pressure. Over the past 2 years, the market share of Korean manufacturers in the European Union has fallen from nearly 80% in 2022 to 60% in 2024, to the benefit of Chinese manufacturers. A key reason for this is the increased success of the LFP chemistry. In contrast, the share of Korean companies in the United States grew from about 20% in 2022 to over 35% in 2024, at the expense of Japanese manufacturers.

Korean manufacturers are rising to the challenge and are now [investing](#) in LFP, including through [innovation](#) efforts, and are scaling up LFP battery production, including in [Europe](#) and the [United States](#). LFP battery development is also advancing in Japan: In early 2025, Nissan [announced](#) plans to build an LFP battery plant after receiving [government certification](#) in late 2024. Yet Chinese manufacturers are continuing to innovate LFP [chemistry](#) and [performance](#) simultaneously, further raising the bar for other producers. At the same time, the Chinese government's [proposed](#) export restrictions on [advanced](#) LFP technologies could limit technology transfer.

In the United States, Panasonic – historically Tesla's main battery partner – remained the country's largest battery producer in 2024, accounting for nearly half of the electric car market. However, its share is declining as Korean producers like LG Energy Solution, Samsung and SKI invest heavily in the US market. Tesla also ramped up its own battery production in 2024, primarily for use in the Cybertruck.

Battery production in the United States is expanding rapidly, spurred by the implementation of [tax credits](#). Manufacturing capacity has doubled since 2022, reaching more than 200 GWh in 2024, with nearly 700 GWh of [additional capacity](#) under construction. Around 40% of existing and committed capacity is being operated or developed by established battery makers in close collaboration with automakers. Nevertheless, the cost of production remains [higher](#) than in Asia, if not accounting for incentives, and sudden policy shifts could affect the emerging US battery industry, [increase](#) production costs, or slow expansion in the near to medium term.

Share of electric car battery sales by battery manufacturer's headquarters, 2022-2024



IEA. CC BY 4.0.

Notes: Battery refers to battery cells. It reflects the batteries installed in vehicles sold in each region, and not the battery demand for vehicles manufactured in each region. Electric car and battery stockpiling are excluded from the analysis.

Sources: IEA analysis based on data from [EV Volumes](#).

Battery innovation is accelerating, and it is not limited to new chemistries

Advanced battery technologies under development include solid-state, sodium-ion, lithium-sulphur, iron-air, and redox-flow batteries, among others. Some of them, like iron-air and redox-flow batteries, target different applications to established lithium-ion technologies, such as longer-duration storage for grid applications. Others, like sodium-ion batteries, aim to reduce dependence on lithium. Lastly, technologies like solid-state and lithium-sulphur batteries could also accelerate electrification in sectors that require or would benefit from higher energy densities, such as long-haul electric trucks or short-haul boats and planes. However, their deployment in these sectors will depend on meeting stringent [safety requirements](#) and on their [total cost of ownership](#).

Sodium-ion batteries gained significant attention in 2022 as lithium prices surged, leading to the [first](#) EVs using the technology. Despite enthusiasm waxing and waning as a result of material supply chain challenges and falling lithium prices in 2023 and 2024, CATL – the world’s largest battery producer – announced its [second generation](#) of sodium-ion batteries in 2025, alongside the launch of a dedicated sodium-ion battery [brand](#). Meanwhile, BYD is also investing in sodium-ion battery production for [EVs](#) and [battery storage](#). In March 2025, HiNa launched its [new](#) sodium-ion battery, which offers improved energy density and faster charging compared to the previous generation, heralding a promising year for this technology. However, recent [analyses](#) indicate that sodium-ion batteries will require either increased energy density or more favourable operating conditions – particularly higher lithium prices – to compete with LFP batteries on a price per kWh basis. Nevertheless, sodium-ion technologies could play a significant role during times of elevated lithium prices and may offer a cheaper option for batteries in cold climates, where LFP batteries typically perform [less well](#).

In 2024, **solid-state batteries** moved closer to commercial reality with new large prototypes and manufacturing investments from [Samsung SDI](#), [Toyota](#), [NIO](#), [Honda](#), [Quantum Scape](#), [BASQUEVOLT](#), and [Factorial](#), among others, and the creation of a government-led Chinese battery [alliance](#), including large producers such as CATL, BYD, SAIC and Geely, to accelerate solid-state battery development. Despite this, their potential advantages, including enabling higher ranges and safety, still need to be [demonstrated](#) for battery packs manufactured at scale and tested under controlled, realistic and standardised conditions. The technology readiness level (TRL) of solid-state batteries therefore remains at large pilot stage ([TRL 6](#)), although this could change rapidly with companies like [Toyota](#) and [BYD](#) planning first mass production by 2027-2028. However, volumes will be [limited](#) initially, and it will take several years following roll-out for solid-state batteries to eventually become competitive with lithium-ion batteries. Additionally, “solid-state batteries” is often used as a generic term covering a [range](#) of options between fully solid-state and incumbent lithium-ion batteries, which creates some confusion. The first “solid-state batteries” to be commercialised might be semi- or quasi-solid-state batteries – for example using gel-like electrolyte² or incorporating small volumes of liquid electrolytes – as they could help address some scale-up [challenges](#) and reduce production costs.

Lithium-sulphur (Li-S) batteries, promising higher specific energy (Wh/kg) and lower reliance on critical minerals, have also gained momentum. The US start-up Lyten [announced](#) the world’s first Li-S gigafactory, while Stellantis [partnered](#) with Zeta to commercialise this technology by 2030. However, several challenges remain, including improving volumetric energy density (Wh/L), enhancing

² French company Blue Solutions already [markets](#) a semi-solid-state battery using lithium metal anode and a gel electrolyte that requires the battery to be heated during use, and [plans](#) to invest more than EUR 2 billion in a factory in France by 2030.

durability, and addressing safety concerns related to the use of lithium metal anodes. Overcoming these hurdles will be key to enabling real-world applications.

Advances in battery **recycling** are also being made, for example [combining](#) pyrometallurgy and hydrometallurgy processes, [increasing](#) the recovery efficiency of reactive metals such as lithium, recycling the [graphite](#) in the anodes, or through the development of new recycling approaches, such as [electrochemical extraction](#). However, recycling feedstock availability remains limited, restricting the impact of recycling in the short term (see Box [below](#)).

Options for **battery reuse** at the end of their first life for less demanding applications are also under development, with the [majority](#) of companies in the sector based in Europe. However, the economic viability of reuse is complicated by upstream competition from recyclers looking for access to end-of-life batteries, downstream competition from the falling prices of new batteries, and [challenges](#) associated with battery dismantling and repurposing while ensuring strict safety standards. [Second-hand](#) EV markets may play a larger role in supporting reuse and also increasing [affordability](#).

Innovation extends far beyond battery chemistries, and the already broad landscape of battery innovation is getting even broader.³ In 2023 and 2024, there was a remarkable surge in **improvements for incumbent lithium-ion batteries**, from [superfast charging](#) and [“no-degradation”](#) batteries to [ultra-energy-dense](#) batteries and new [charging platforms](#), [manufacturing processes](#), [cell formats](#) and [pack designs](#), among others. Advances in manufacturing are also notable: for example, [artificial intelligence](#) for image analysis can enable the [early detection](#) of battery defects and their root causes, thereby improving production yields and reducing scrap rates. This capability is critical for scaling up production given the pace of modern gigafactories – a 50 GWh facility can produce up to 10 million (cylindrical) or hundreds of thousands (prismatic) EV battery cells per day.⁴ The nature and pace of innovations in legacy technologies are already making a big impact on the market, posing a formidable challenge for emerging technologies to compete.

³ See IEA (2025) [State of Energy Innovation](#) for deeper insights into how this landscape is evolving.

⁴ Assuming an average plant utilisation factor of 85% over the year, a cell voltage of 4 volts, and a cell capacity of 60 ampere hours (prismatic) and 3 ampere hours (cylindrical).

Battery recycling is important, but its benefits will take time to materialise

Battery recycling is crucial for the battery industry's long-term sustainability, and is being promoted by industry organisations and governments through dedicated [investments](#) and regulations. For instance, the EU [Batteries Regulation](#) sets requirements for minimum recycling efficiency and recycled contents that will increase progressively. Recycled battery metals such as nickel, cobalt and lithium can incur [80% less](#) GHG emissions than primary materials produced from mining and refining, while also [reducing](#) local environmental impacts, including water use and biodiversity loss near mines.

In addition to sustainability, governments are prioritising supply chain security. Recycling can enhance resilience for consumer countries without direct access to battery minerals, but only if there is a simultaneous development of the midstream battery supply chain, particularly the cathode active material (CAM) industry. To realise the security benefits, end-of-life batteries must be recycled, and the recovered minerals processed and utilised domestically, or in a partner country, to produce CAMs for batteries. However, the CAM industry is progressing more [slowly](#) than the battery industry in Europe and the United States. In the absence of a domestic industry, the recycled minerals would need to be exported for processing and then reimported as CAMs, or within batteries or EVs, thereby forfeiting the supply chain security advantage.

Recycling industries in Europe and North America have focused on NMC batteries – the [main](#) chemistry used for electric cars in these regions, but [advances](#) in LFP batteries are now making them more attractive outside of China, where LFP already leads the EV battery market. The rise of [cheaper](#) batteries such as LFP poses a [challenge](#) to the economic viability of battery recycling, which requires sufficiently high concentrations of valuable minerals in batteries to be profitable. Nevertheless, this does not preclude LFP battery recyclability, and NMC recycling facilities can be modified to process LFP batteries, though this might require different business models. [Toll-based models](#) – through the implementation of [gate fees](#) – can ensure the economic viability of LFP recycling, especially when lithium prices are low. Regulations would need to define how these toll fees are distributed to guarantee the proper management of all end-of-life batteries.

It will take time for the battery recycling industry to significantly affect primary mineral demand, as feedstock remains limited and EV batteries may last [longer](#) than previously expected. End-of-life EV and storage batteries are expected to account for only [one-third](#) of recyclable feedstock by 2030 and even less in the interim, with manufacturing scraps making up the rest. It will require about a decade for end-of-life batteries to become the main feedstock. In 2035, if countries fulfil their announced climate pledges, recycling would [reduce](#) global primary demand for cobalt by over 15% (40% in 2050), and for lithium and nickel by only about 5% (25% in 2050).

Battery production and trade

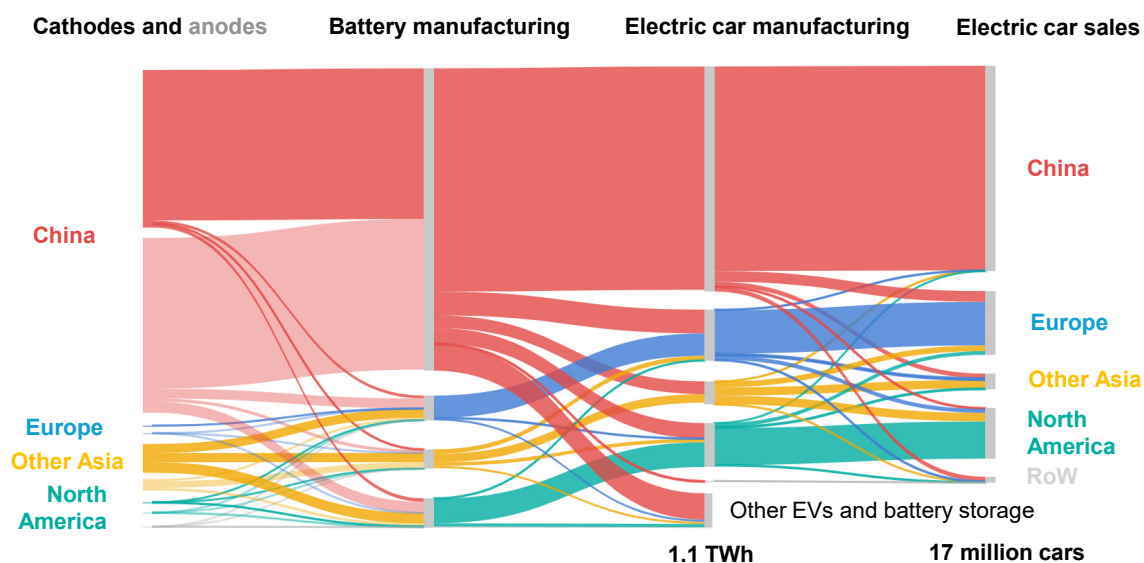
The battery market is being reshaped by two distinct trends: increasing consolidation, and government-led efforts to geographically diversify battery supply chains. Whereas markets used to be regionalised and small, they are now global and very large, and uncertainty over technological development is giving way to standardisation. In this [new phase](#), economies of scale, partnerships along the supply chain, manufacturing efficiency, and the capacity to bring innovations swiftly to market will be even more crucial for manufacturers to remain competitive.

The battery supply chain remains geographically concentrated

The EV and battery supply chain becomes increasingly geographically concentrated when moving upstream from car manufacturing to battery cell and component production. China was responsible for 80% of global battery cell production in 2024, while the remainder was produced in the United States, the European Union, Korea and Japan. Importantly, the production of lithium-ion battery manufacturing [equipment](#) is also highly concentrated, with China, Korea, and Japan [leading](#) the market. China has also established a near monopoly on battery components production, supplying almost 85% of cathode active materials – including NMC and LFP chemistries – and over 90% of anode active material production, predominantly graphite. Outside of China, only Korea and Japan offer sizeable production capacity for cathode components. Korea also produces anodes, and Indonesia is [expected](#) to bring some diversification to the market. Nonetheless, China is set to remain the largest producer of batteries and their components by some distance in the medium term, based on announced projects and [competitive advantages](#).

The [geographical concentration](#) of [battery mineral](#) mining and refining needed for battery cathodes and anodes is also a concern. In 2023, Australia, Chile and China mined about 85% of global lithium, with almost 65% refined in China and another 25% in Chile. Indonesia accounted for over half of nickel mining in 2023, while China and Indonesia together refined more than 60%. In the same year, the Democratic Republic of Congo was home to almost two-thirds of the world's cobalt mining, though three-quarters of all cobalt refining was handled in China. Graphite supply, the only critical mineral used for anodes today, is even more concentrated, with China responsible for 80% of mining and over 90% of refining.

Global manufacturing and trade flows of electric cars, lithium-ion batteries, and key components, 2024



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Notes: RoW = Rest of World; EV = electric vehicle; Cathodes and anodes refer to cathode and anode active materials. Flows are normalised to the battery (cell) manufacturing step, with cathode and anode active materials normalised such that their sum is scaled to the battery cell volume. Numbers below the charts refer to the total demand, not only the traded volume. The lighter-colour version of the flows going to battery manufacturing represents the anodes (anode active materials). Battery applications different from EVs and battery storage are excluded from the analysis. Electric vehicle and battery stockpiling are excluded from the analysis.

Sources: IEA analysis based on [EV Volumes](#), [Benchmark Mineral Intelligence](#), and [Bloomberg New Energy Finance](#).

Battery manufacturing capacity continues to grow

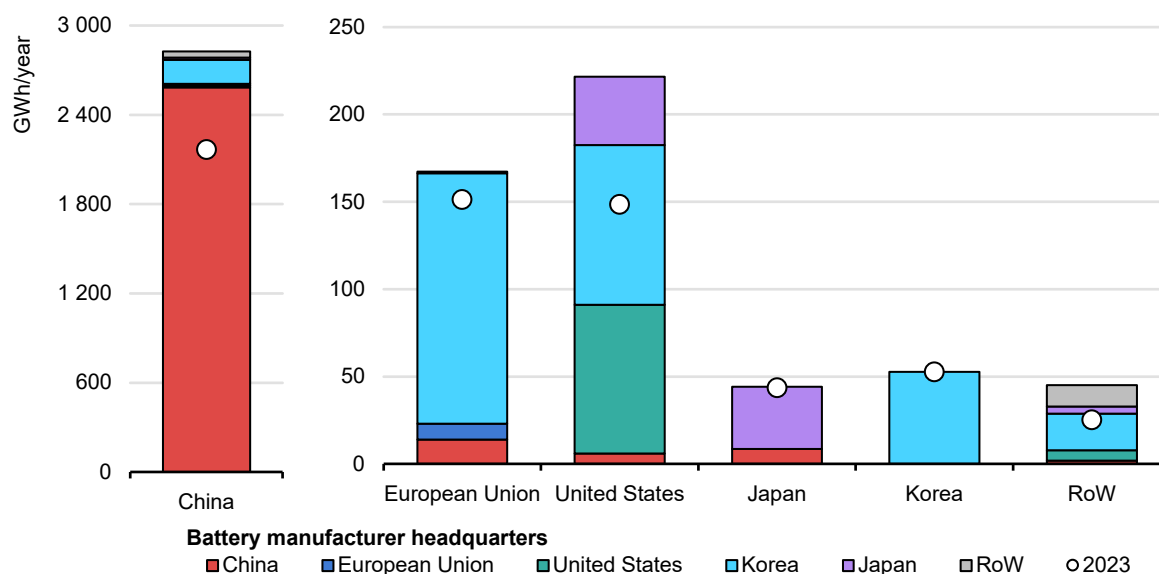
Global battery (cell) manufacturing capacity grew almost 30% in 2024 to reach more than 3 TWh – three times EV and battery storage demand in the same year. About 85% of global manufacturing capacity is in China, showing little change from 2023, and over 75% is owned by Chinese producers. Manufacturing capacity in the United States grew by almost 50%, led by Korean companies [attracted](#) by [tax credits](#), which accounted for nearly 70% of the growth in 2024. This led installed capacity in the United States to surpass that in the European Union, which nonetheless increased by 10% in 2024 despite the Northvolt plant in Sweden being halted following its [bankruptcy](#). The first [Indian](#) and [Indonesian](#) battery plants also opened in 2024, totalling more than 5 GWh/year and 10 GWh/year of manufacturing capacity, respectively.

Korean manufacturers remained the largest investors in overseas battery manufacturing capacity,⁵ accounting for over 400 GWh globally, compared to 60 GWh for Japanese and 30 GWh for Chinese producers. Korean manufacturers, such as LG Energy Solution (with capacity in Poland), and

⁵ This refers solely to existing manufacturing capacity as of 2024, and it accounts only for factories located outside the home country of the battery producer's headquarters.

Samsung and SKI (in Hungary), continue to be the main source of capacity in the European Union, and their importance is growing in the United States. If all announced projects are completed in full, the manufacturing capacity of Korean manufacturers outside of Korea would reach more than 1.1 TWh by 2030, 85% more than the announced overseas manufacturing capacity of Chinese battery producers.

Installed lithium-ion battery cell nameplate manufacturing capacity by region and location of manufacturer's headquarters, 2024



IEA. CC BY 4.0.

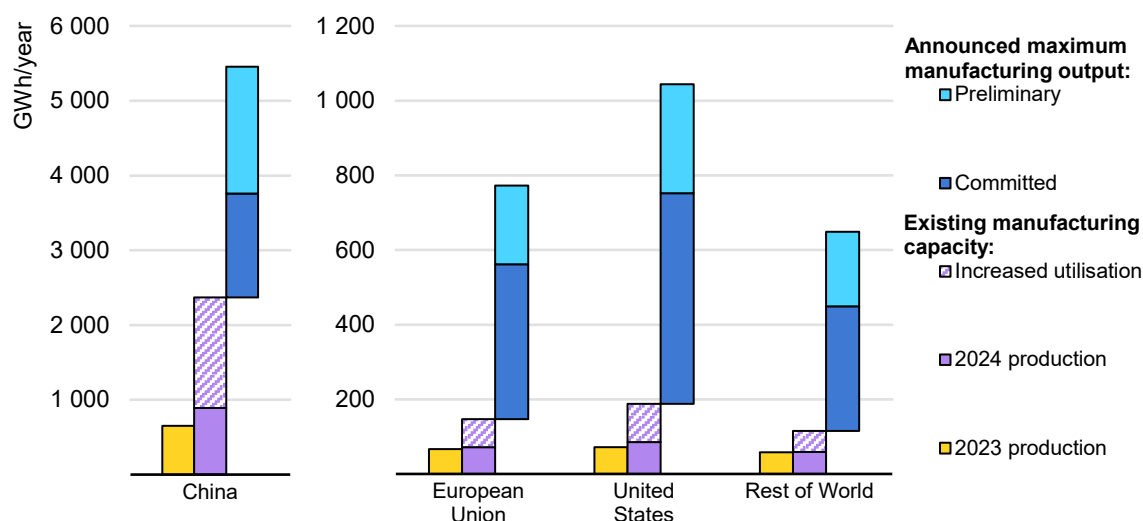
Notes: RoW = Rest of World. Lithium-ion battery manufacturing capacity installed in China, European Union, United States, Japan, Korea, and other countries sorted as a function of the company headquarters location (colours). For companies headquartered in a country but owned by companies headquartered in a second country, the headquarters of the owning company is considered for this analysis. The manufacturing capacity of joint ventures between automakers and battery producers is classified according to the battery producer headquarters.

Source: IEA analysis based on data from [Benchmark Mineral Intelligence](#).

Asian manufacturers are leading battery market expansion

Expansion plans for manufacturing capacity can drive geographical diversification in the battery industry. Committed projects – i.e. those that are either under construction or have reached a final investment decision – would increase manufacturing capacity in China by nearly 60%, and almost quadruple capacity in the European Union and the United States. This expansion would lead global manufacturing capacity to grow to about 6.5 TWh (and more than 9 TWh if accounting for all announcements), up from 3.3 TWh in 2024, and reduce China's share of global manufacturing capacity to about two-thirds by 2030, down from 85% in 2024. Ownership distribution is expected to diversify less, given that capacity expansion is predominantly driven by established manufacturers headquartered in China, Korea or Japan, whose expertise in the battery sector provides a significant competitive edge.

Historical production and announced expansion of battery manufacturing maximum output by region, 2023, 2024 and 2030



IEA. CC BY 4.0.

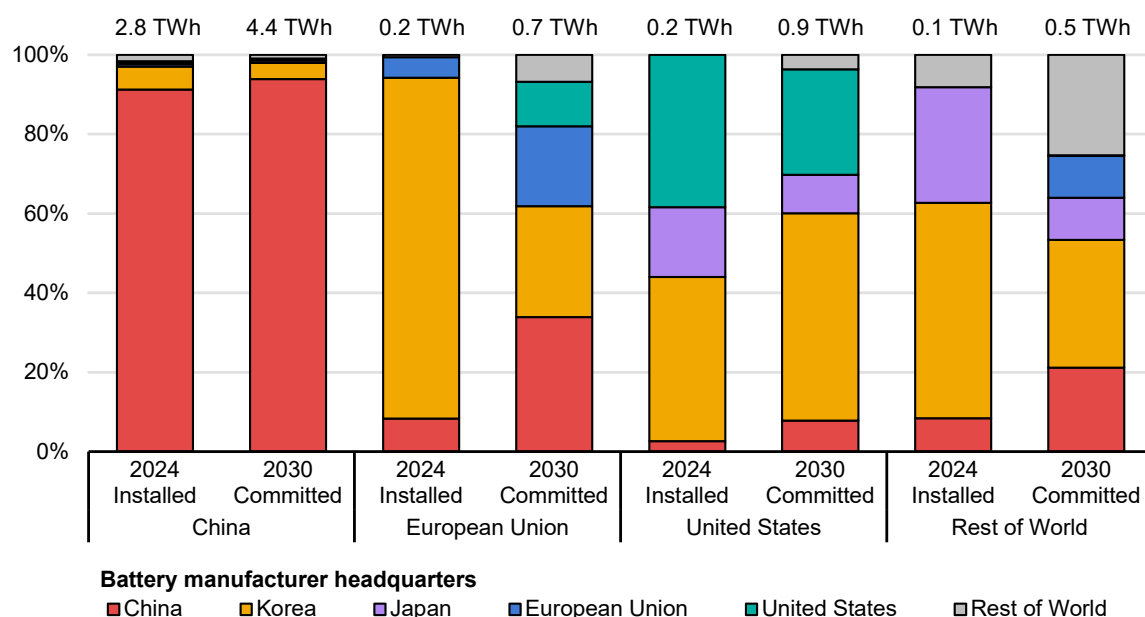
Notes: Increased utilisation refers to the gap between 2023 production levels and existing capacity being utilised at 85%, which is assumed to be the maximum average utilisation rate. A utilisation rate of 85% is also used for both committed and preliminary manufacturing capacity in 2030. Manufacturing capacity refers to companies already certified to serve both the EV and stationary storage markets and companies not yet certified to serve the EV market. Committed refers to plants that have reached a final investment decision and are starting or have already started construction works, and preliminary to plants that have been announced but are not yet being built. Production refers to EV battery and battery storage, and EV and battery stockpiling are excluded from the analysis. Data to end of Q1 2025.

Sources: IEA analysis based on data from [Benchmark Mineral Intelligence](#), [Bloomberg New Energy Finance](#), and [EV Volumes](#).

The European Union is the largest single destination for [overseas investments](#) by Chinese battery producers, whose share of manufacturing capacity in the region could quadruple, rising from less than 10% in 2024 to more than 30% by 2030. These investments are an opportunity for technology transfer, bringing the expertise needed to scale up production in the region and [drive down](#) costs, which would benefit domestic automakers at a critical time for the industry. However, this may also create challenges for the region's established and emerging battery producers.

The share of battery manufacturing capacity in the European Union owned by Korean producers is expected to fall sharply from about 85% in 2024 to nearly 30% in 2030, while the share of EU-based companies could reach 20%, up from 5% at the end of 2024. However, this includes plants facing significant uncertainty, such as the Northvolt plant in [Germany](#), for which Germany assumed over [EUR 600 million](#) in debt, and which is [not](#) directly affected Northvolt's [bankruptcy](#). More generally, [concerns](#) are growing about the ability of smaller European producers to [scale up](#) production and compete with established global players, which may lead to a much smaller share of the future EU battery market being captured by domestic manufacturers.

Share of nameplate manufacturing capacity by region and location of battery producer's headquarters, 2024-2030



IEA. CC BY 4.0.

Notes: Committed refers to the sum of the installed manufacturing capacity (2024) and plants that have reached a final investment decision and are starting or have already started construction works. Manufacturing capacity refers to battery cells. For companies headquartered in a country but owned by companies headquartered in a second country, the headquarters of the owning company is considered for this analysis. The manufacturing capacity of joint ventures between automakers and battery producers is classified according to the battery producer headquarters. Numbers above each bar indicate the total nameplate manufacturing capacity in terawatt-hours (TWh) of batteries per year. Data to end of Q1 2025. Sources: IEA analysis based on data from [Benchmark Mineral Intelligence](#) and [Bloomberg New Energy Finance](#).

Collaboration between Korean or Japanese battery producers and OEMs operating in the United States is proving successful, with Korean companies leading investments in the country. They held or participated in 40% of the battery manufacturing capacity in the United States in 2024, and committed investments are expected to boost their share to over 50% by 2030. Meanwhile, based on committed projects, the share of Japanese companies would fall by almost half by 2030, and that of domestic companies like Tesla would drop from nearly 40% in 2024 to less than 30% in 2030. In addition, some companies primarily or fully owned by Chinese groups, such as Envision and Gotion, have also invested or announced plans to invest in the United States, but recent [policies](#) might lead to the cancellation of these plans.

Outside of today's three main EV markets, 60% of committed capacity is set to be added in other advanced economies thanks to growing demand and government support, including in [Canada](#), other [European](#) countries, [Korea](#), and [Japan](#). The remaining more than 150 GWh of committed manufacturing capacity is being built in Southeast Asia, India and Morocco. Although these regions have attracted fewer investments to date due to limited domestic battery demand, they are increasingly generating interest from battery manufacturers.

Southeast Asia is attracting significant Chinese investments, which could speed up technology and innovation transfer. Indonesia, home to half the world's mined nickel needed for NMC batteries, is also investing heavily in battery component production – such as cathode and anode active materials – and its first [graphite anode](#) plants began production in 2024. India also has the potential to [unlock](#) a substantial battery market and is [investing](#) in domestic battery production, but realising its ambition to become a major battery manufacturer will require [additional investments](#) and clear policy signals supporting EV demand. In Morocco, abundant phosphate reserves – a mineral essential for LFP batteries – along with an established car manufacturing industry and free trade agreements with the European Union and the United States, have spurred over [USD 15 billion](#) in announced investments. These investments comprise lithium processing and battery and component manufacturing, including a large battery manufacturing plant of [100 GWh](#), the first in Africa.